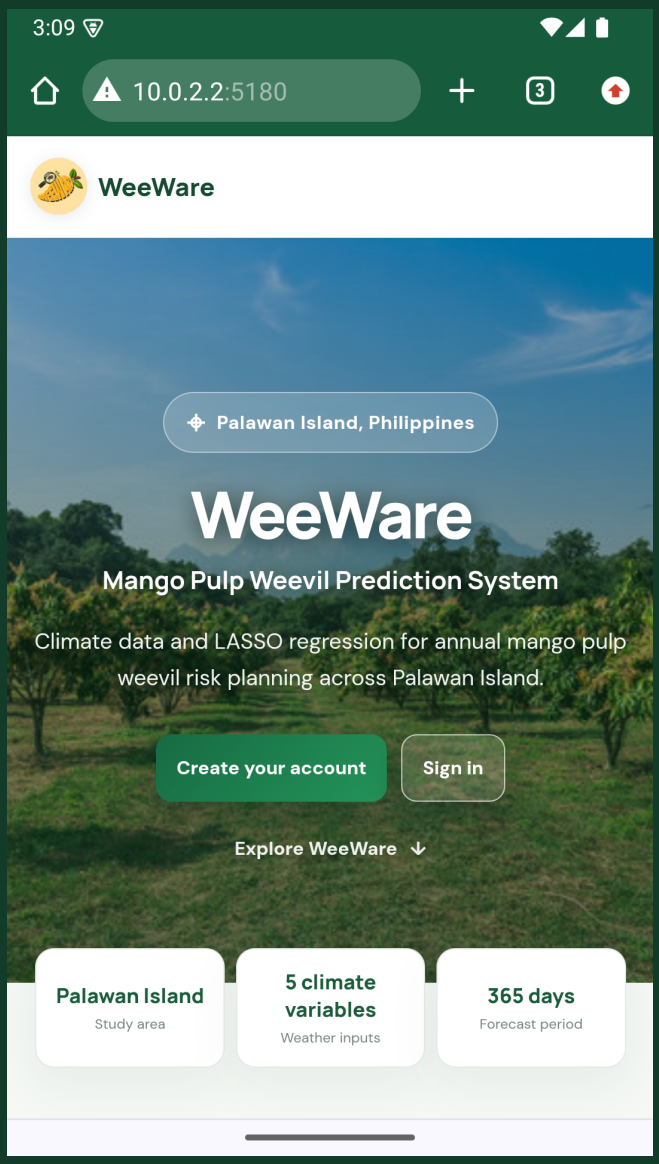
WEEWARE | SYSTEM GUIDE

End-to-end system documentation

Pages, prediction flow, API, services, and user controls



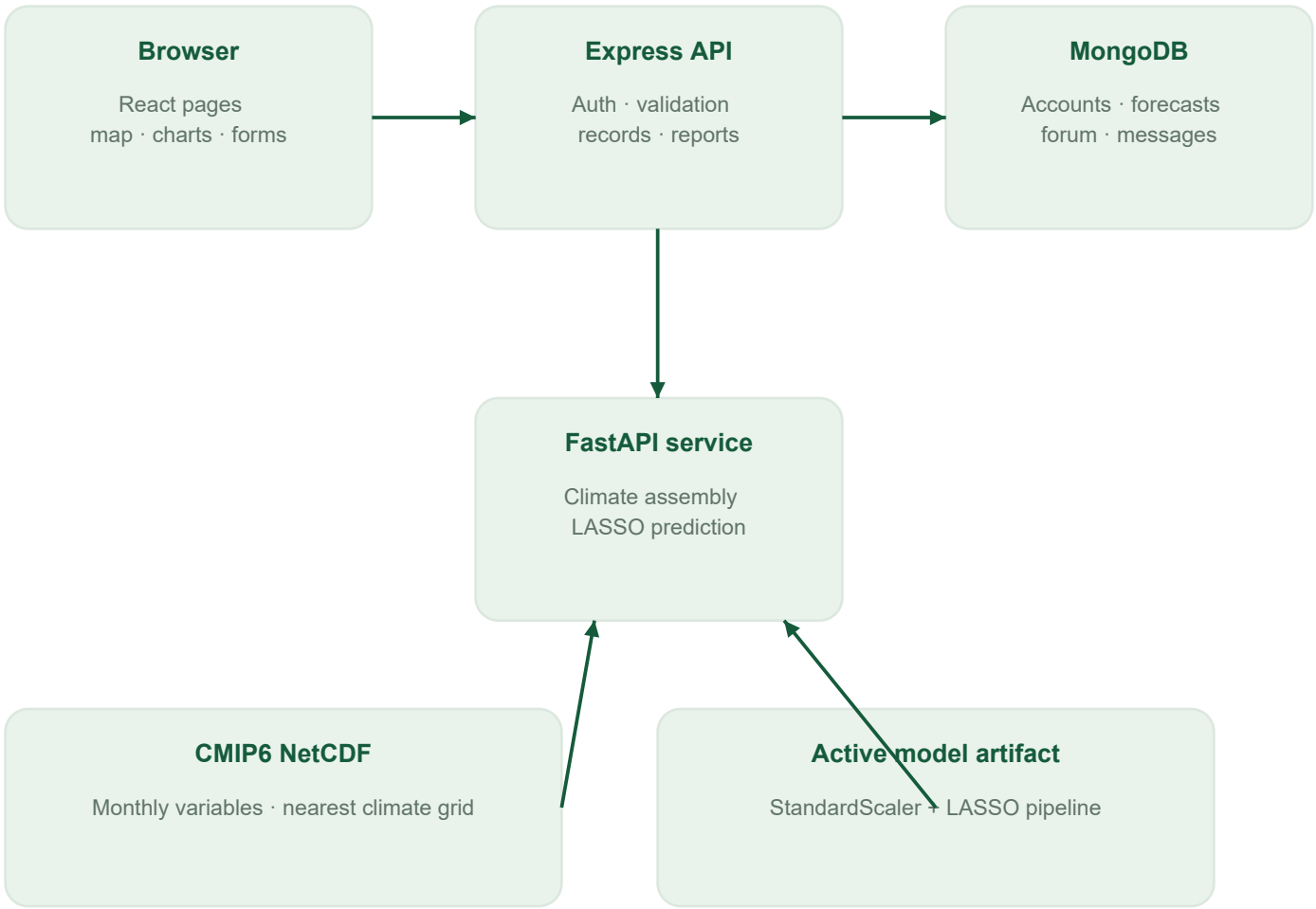
Mango pulp weevil monitoring across Palawan

Prepared 11 November 2026 | Repository version 2.0

01 / ARCHITECTURE

How the pieces fit together

WeeWare is a responsive React application for reviewing annual climate patterns and producing a climate-informed mango pulp weevil screening estimate. A normal forecast crosses four service boundaries: the browser, Express API, MongoDB, and the Python climate/model service.



Local development addresses

Service	Local default	What it does
Vite / React	localhost:5180	Mobile/desktop development client; proxies API, uploads, and Socket.IO.
Express API	127.0.0.1:5021	Authentication, validation, persistence, API routes, real-time events.
FastAPI	127.0.0.1:8021	Climate data extraction and model operations.
MongoDB	127.0.0.1:27017	Application records and account-owned history.

A prediction from start to finish

1. Authenticate
Login issues a signed HTTP-only cookie. Subsequent API requests include it automatically; the API resolves the account ID and role from the JWT.

2. Select the map point
The user clicks or drags the pin in the Palawan map. The selected coordinates are passed into weather extraction and prediction. The configured accepted extent is 7.0–12.9°N and 116.0–121.0°E.

3. Choose the date window
The forecast must contain exactly 365 inclusive days. The end date is 364 days after the start, and the start must be no earlier than January 1 of the current year.

4. Review weather
The browser requests the 365-day climate window. The service returns five weather values, monthly summaries, source names, and coverage counts.

5. Run LASSO
Express validates the same window and location again, then passes dates and coordinates to FastAPI. Python assembles the ordered feature vector, applies the active StandardScaler/LASSO pipeline, and produces the estimate and risk band.

6. Persist and act
Express stores the input values, source metadata, output, and model information. The user can save the report, download its PDF, revisit history, or contact an agriculture account.

The output is a screening estimate. It is not a calibrated probability or a farm-level diagnosis; field inspection and local agriculture guidance remain important.

How climate values are assembled

The active extraction logic is in `ml-service/climate.py`. It opens the imported monthly NetCDF files with xarray, finds the latitude/longitude coordinate names, selects the nearest grid point, filters the requested year/month, converts units when needed, and preserves the source filename.

Feature	CMIP6 variable	Model unit	Conversion / aggregation
Rainfall	pr	mm/year	Flux: multiply by seconds/day and days in month. mm/day: multiply by days in month. Sum represented days.
Temperature	tas	°C	When input is Kelvin, subtract 273.15. Day-weighted average over the date window.
Relative humidity	hurs	%	Use source values when present; day-weighted average.
Solar radiation	rsds	MJ/m²/day	W/m² × 0.0864; day-weighted average.
Wind speed	sfcWind	m/s	Use source values when present; day-weighted average.

Data fallback order

For each feature/month, use an available original annual weather row when one matches; otherwise read a covering CMIP6 monthly series at the selected coordinate. Missing coverage falls back to a labeled training-climatology value. The API reports each source so mixed/fallback inputs remain visible.

Window aggregation

The selected dates are traversed day by day. Monthly input is reused for days within that month. Rainfall is accumulated across the 365 days; temperature, humidity, solar radiation, and wind are averaged with day counts as weights.

Climate file coverage

The archive has seven imported monthly files. It mixes historical, SSP2-4.5, and decadal-forecast sources; the fields do not make one complete, consistent future climate series. NetCDF grid cells are coarse, so the nearest-grid result is an area estimate, not a farm weather station.

Model, training, and result meaning

Inputs and artifact

The five model inputs are annual rainfall (mm/year), temperature (°C), humidity (%), solar radiation (MJ/m²/day), and wind speed (m/s). The active artifact is a scikit-learn StandardScaler + LASSO regression pipeline. The documented Palawan retraining used alpha 0.05, 100 annual location rows, 23 locations, and years 2014–2025.

Prediction output

The active model uses a 0–100 infestation-rate scale. Results are constrained to this configured range. Low is below 33%; Medium is 33% to below 66%; High is 66% or higher. The UI also shows suggested monitoring actions, the selected dates/location, and the extracted weather context.

Validation evidence

Leave-one-year-out validation holds out all locations from one year at a time, fits on the other years, and compares predictions with held-out labels. The recorded MAE is 15.00 percentage points; the simple training-mean baseline is 17.73. The held-out-location LASSO MAE is 17.76. This is exploratory evidence, not a production guarantee.

Interpretation limits

The dataset is modest; some labels repeat across years and survey definitions differ. Weather rows approximate municipality centers rather than farms. Do not interpret the output as a calibrated probability, causal effect, or field diagnosis. See `docs/model-validation.md` and `docs/data-inventory.md` for experiment and source detail.

Password reset: delivery flow and root cause

The forgot-password page posts the account email to `POST /api/auth/password/forgot`. For privacy, unknown addresses receive the same generic response as known addresses. For a known account, Express generates a cryptographically random one-time token, stores only its SHA-256 hash with a 30-minute expiry, builds a link from `FRONTEND_URL`, and asks Nodemailer to send it through the configured SMTP server.

Root cause found in the application code

When SMTP sending failed, the route logged the exception but still returned HTTP 200 with the generic “check your inbox” message. The browser therefore showed success even when no reset email had been sent. The route now checks SMTP availability and validates the reset-page URL before looking up an account; a service/configuration failure returns the same HTTP 503 for every address. If a send fails after those checks, the route removes the failed token or restores the previous valid challenge and keeps the account response generic to prevent email enumeration.

What must be configured for delivery

This checkout has no `backend/.env` and no SMTP credentials. The deployed API must have valid SMTP_HOST, SMTP_PORT, SMTP_SECURE, SMTP_USER, SMTP_PASS, SMTP_FROM_EMAIL, and FRONTEND_URL values in its protected runtime environment. FRONTEND_URL must be the public website origin so the emailed link returns to the correct reset page. Code changes alone cannot deliver email without a working provider account.

How to interpret the status

The same successful API response covers an unknown address and a request where no error was returned; this avoids exposing registered emails. It does not guarantee that a message reached the inbox. A 503 means the SMTP transport failed its availability check before any account lookup. Check backend logs for later send errors and verify provider credentials, sender authorization, TLS/port settings, DNS, and outbound network access. Never put SMTP credentials in GitHub or screenshots.

Reset token controls

Tokens are random, hashed before storage, single use, and expire after 30 minutes. Resends are limited to one request every 30 seconds. Reset requests continue to return the same response for unknown email addresses.

Page directory

Route	Page	Purpose
/	Home	Project overview, Palawan scope, and account entry.
/login	Sign in	Authenticate an existing account.
/signup	Create account	Profile, consent, and email verification.
/forgot-password	Password recovery	Request a secure reset link.
/reset-password	Password reset	Set a new password from a valid link.
/app/prediction	Prediction	Location → dates → weather → result.
/app/history	History	Review, save, download, or delete forecasts.
/app/saved	Saved reports	Reopen/delete saved forecasts.
/app/forum	Community forum	Search, post media, like, comment, and reply.
/app/profile/:id	Member profile	Public member information and message link.
/app/messages	Messages	Direct, designated group, and admin broadcast chats.
/app/ai	AI assistant	Saved assistant conversations, if provider is configured.
/app/settings	Settings	Profile, privacy, language, password, account.
/app/system-iteration	Researcher	Model, datasets, experiments, retraining.
/app/da	DA dashboard	Forecast and risk summaries for DA/admin.
/app/admin	Admin	Users, moderation, services, traffic, configuration.

Legacy map, calendar, and climate URLs redirect into the current prediction flow; the
router.

API reference: Express

The Express service is mounted at `/api`. Authentication is cookie-based for the browser; route middleware applies account and role checks. Public health is available at `/api/health`. Express does not currently publish its own Swagger UI.

Route group	Key operations	Access purpose
/auth	Signup, verify, login/logout, current account, password reset.	Public auth actions; current-account route requires login.
/users	Current/public profiles, search, profile/password update, deletion.	Personal changes require login; public profile reads may be anonymous.
/climate	Years, year detail, 365-day window.	Climate proxy endpoints.
/predictions	Create/list/get/delete, save, PDF report.	Signed-in user; history is scoped to that user.
/saved-predictions	List and delete saved items.	Signed-in owner.
/posts + /comments	Search/list/create, media, comments/replies/likes, update/delete.	Public reads where configured; writing requires login.
/messages	Threads, direct/group messages, chunked group files, broadcast.	Signed-in; group membership/admin restrictions apply.
/uploads	Media upload and limits.	Upload requires login.
/notifications	List/unread count and read-state updates.	Signed-in account.
/ai/conversations	List/create, append message/response.	Signed-in roles.
/ml	Model info, preview/train, experiments, activation.	Researcher/admin.
/moderation	Removal requests and moderation queue/actions.	Member request; admin review/actions.
/system	Overview, accounts/roles, schema, config, health, traffic; DA forecast summary.	Admin, except DA/admin forecast summary.

API reference: climate and LASSO

Method + path	Purpose
GET /health	Service/model and CMIP6-file health summary.
GET /model/info	Active model metadata/configuration.
GET /train/status	Current retraining state.
GET /train/experiments	List saved training experiments.
GET /train/template	Download a training input template.
POST /train/preview	Validate a CSV/XLSX dataset without activation.
POST /train	Train and validate a supplied dataset.
POST /train/palawan	Train from the prepared Palawan dataset.
POST /train/experiments/{id}/activate	Activate an existing experiment.
GET /train/experiments/{id}/dataset	Download an experiment dataset.
DELETE /train/experiments/{id}	Delete an experiment.
GET /climate/years and /year/{year}	Climate year list/details.
GET /climate/window	365-day coordinate/date feature extraction.
POST /predict	Climate-window features through active LASSO inference.
GET /docs	Interactive OpenAPI reference (FastAPI).

The ML service should normally be reached through the Express API. Training/act
stay behind researcher/admin authorization; do not publish the ML port to untruste

Services, storage, and events

Component	Responsibility	Stored state / notes
React + Vite	Pages, form state, maps, charts, API calls.	Client assets; Vite proxies API/uploads/Socket.IO in development.
Express + Socket.IO	JWT auth, role controls, validation, record operations, PDFs, real-time messages/events.	Serves built frontend in production when `frontend/dist` is present.
MongoDB + Mongoose	Accounts and application records.	Users, predictions, saved reports, posts, comments, notifications, messages, conversations.
FastAPI climate/model	CMIP6 reading, fallbacks, inference, training operations.	Seven NetCDF monthly inputs and joblib experiment/model artifacts on disk.
File storage	Community media and group attachments.	Configured upload paths; group transfer uses 8 MiB chunks and up to 1 GiB.
SMTP mailer	Verification/reset messages.	Requires SMTP credentials in runtime secret configuration.
Optional Ollama endpoint	AI assistant responses.	Requires configured URL and model; page may be unavailable without it.

Realtime path

Socket.IO authenticates a socket from the signed cookie/token and joins account/admin rooms. The API emits events for notifications, forum and messaging updates so connected pages can refresh without polling every interaction.

Persistence map

MongoDB stores application records. The active climate reader loads source NetCDF and training CSV from the repository data tree; scikit-learn/joblib loads the active estimator from `ml-service/models/`. Uploaded files use local configured directories, so backups must cover both database and file storage.

Controls, roles, and validation

Control	What it enforces
Signed-in cookie	HTTP-only JWT cookie is checked by `requireAuth`; a bearer token is also recognized by the middleware.
Roles	Farmer/member pages; researcher/admin model workspace; DA/admin forecast dashboard; admin-only system and moderation controls.
Verification	Six-digit email code in six separate boxes; 10-minute expiry, five failed attempts, 30-second resend cooldown.
Forecast request	Exactly 365 days, start on/after current-year Jan 1, required dates, accepted coordinate extent.
Research input	CSV/XLSX only, maximum 12 MiB; preview/validation endpoint is separate from model training.
Media	Images/video for posts, up to eight files; group files stream in chunks, retain original filenames, and cap at 1 GiB.
Account deletion	Confirmation dialog; removes the account and related personal records. Treat as irreversible.

Admin control center

The admin dashboard includes overview, users/roles, forum review, live topology, traffic monitoring, conversation inspection, database schema, system configuration, and health/performance data. Changes are role-gated in Express; the UI is not the security boundary.

Researcher controls

The researcher dashboard can preview a dataset, start a retraining job, use the prepared Palawan set, inspect experiments, download datasets, and activate an experiment. Training routes are protected for researcher/admin.

Privacy and deployment

Keep JWT/SMTP/provider credentials in `.env` or the deployment secret store. Do not commit environment files. Health and service endpoints are intended for the application/operations layer, not public exposure of training controls.

Operating notes and limits

Start locally

- Install Node.js 20+, Python 3.10+, and MongoDB 7+.
- Run `scripts/setup.ps1`; configure `backend/.env` with a strong local JWT secret and MongoDB URI.
- Start MongoDB, then run `scripts/start.ps1` or `npm run dev` from the repository root.
- Open `http://localhost:5180`. The development client proxies `/api`, `/uploads`, and `/socket.io` to Express.

If a page or forecast fails

- Check `/api/health` for Express and the admin health tab for database/model status.
- Check the ML service `/health` and `/docs`; confirm the model artifact and imported NetCDF files are present.
- Confirm start/end dates span 365 days and the pin is within the configured extent.
- Check the returned per-feature source labels. Fallback data can be valid to display while still differing from a complete CMIP6 series.
- If signup/reset email fails, verify SMTP runtime settings; account email actions require a working mail transport.

Interpret with care

The new Palawan model has a 15.00 percentage-point leave-one-year-out MAE and a 17.76-point leave-one-location-out MAE. The simple yearly mean baseline is 17.73 points. The scores are exploratory because the annual dataset is limited, some labels repeat, and weather values are estimated at municipality centers. They are not a guarantee for any farm or future season.

Accuracy limits and 3D analysis

Training facts
The new Palawan model uses 100 annual location rows, 23 labeled locations, and years 2014–2025. It fits a StandardScaler + fixed-alpha LASSO (alpha 0.05) on five annual weather features. The weather values are NASA POWER estimates for municipality centers and the infestation targets come from the supplied records.

Training data versus prediction data
CMIP6 was not used to fit the new LASSO model. Annual NASA POWER municipality-center weather values were paired with the supplied infestation records for training. The live prediction service separately uses available CMIP6 monthly values and labeled fallbacks when its future/coordinate coverage is incomplete.

Main causes of unreliable forecasts
The strongest documented limitations are: few independent observations (100 rows across 23 places and 12 years); repeated infestation labels (65 rows reuse a location label across years); mixed target definitions and aggregation methods (12 earlier annual rows, 36 damage/harvest rows, 52 unweighted sampled-month means); and municipality-center or coarse-grid weather that cannot represent farm conditions. The five weather-only inputs also omit pest monitoring, orchard management, crop stage, and other local factors.

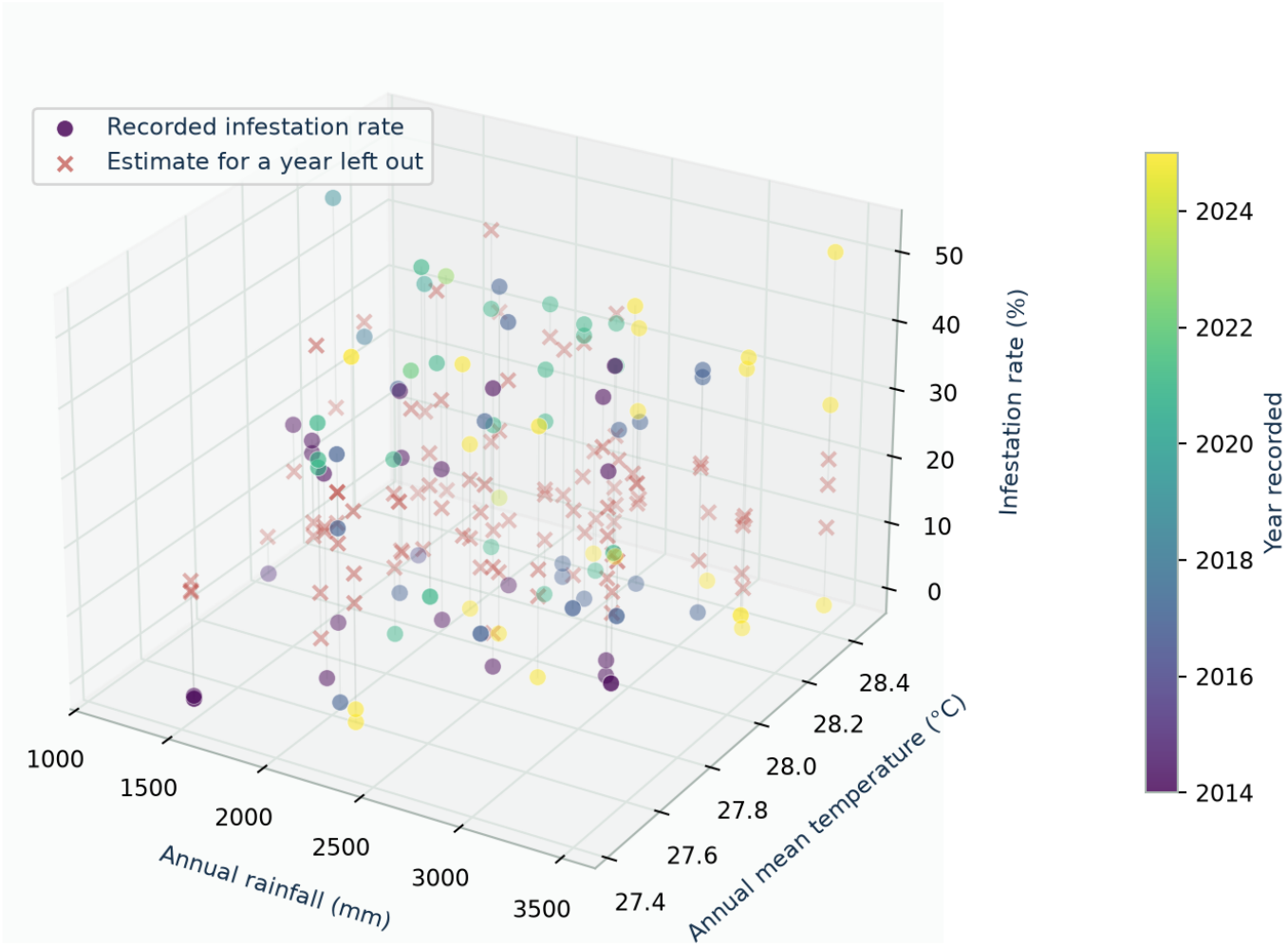
What validation says
When years are held out together, MAE is 15.00 and RMSE is 17.11 percentage points; the training-mean baseline is 17.73 / 18.56. When locations are held out, LASSO MAE/RMSE is 17.76 / 20.41 versus baseline 18.59 / 19.46, so the model has not shown lower large-error risk for unseen locations. This supports limited spatial generalization; it does not identify one single cause.

A note about 30 years
Thirty years is a common climatological-normal period, but it is not a hard requirement in this LASSO training code. More years of independently observed infestation paired with weather would strengthen evaluation; repeating old labels across years does not create the same evidence as new observations.

How to read these 3D figures
The response surface varies rainfall and temperature while holding humidity, solar radiation, and wind at training medians; it is an illustration of the fitted model, not a causal relationship. The validation figures display held-out predictions, not training fit alone. Three dimensions help inspect patterns but do not increase the underlying evidence or make a prediction more accurate.

3D SYSTEM VISUALIZATION

Training view: site-year records and held-out estimates

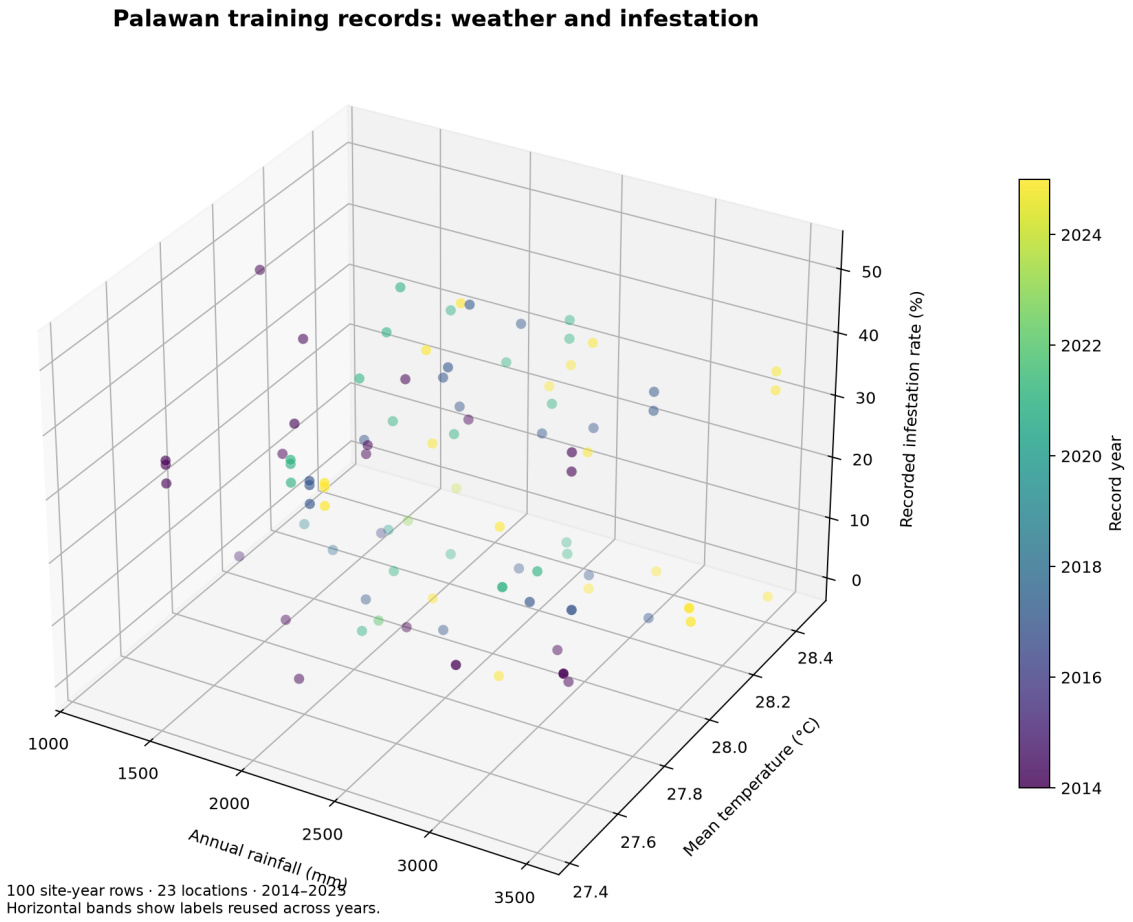


For each red x estimate, the model learned without seeing any entries from that year.

The existing 3D training visualization shows recorded infestation values and estimates made while each year was held out. Each point is a location/year row; 100 rows do not mean 100 independent sites.

3D SYSTEM VISUALIZATION

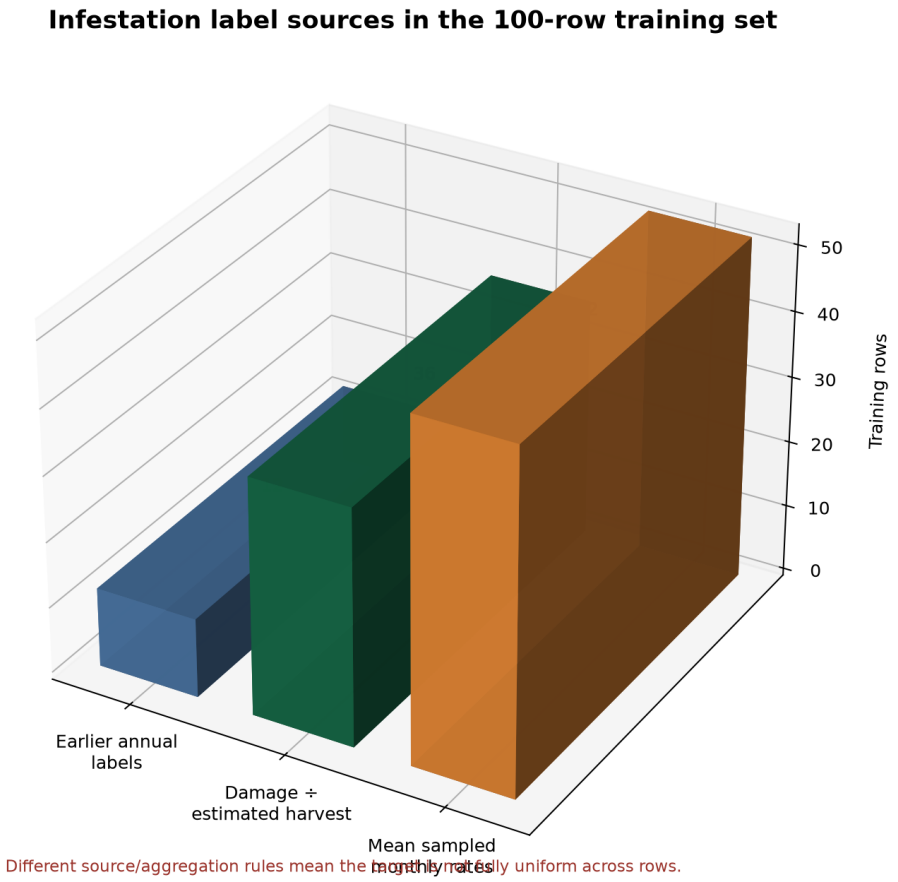
Data view: weather inputs and infestation labels



Each point is one annual location row. Color marks year. Repeated horizontal target bands are consistent with labels reused across years, which reduces independent information for evaluating time trends.

3D SYSTEM VISUALIZATION

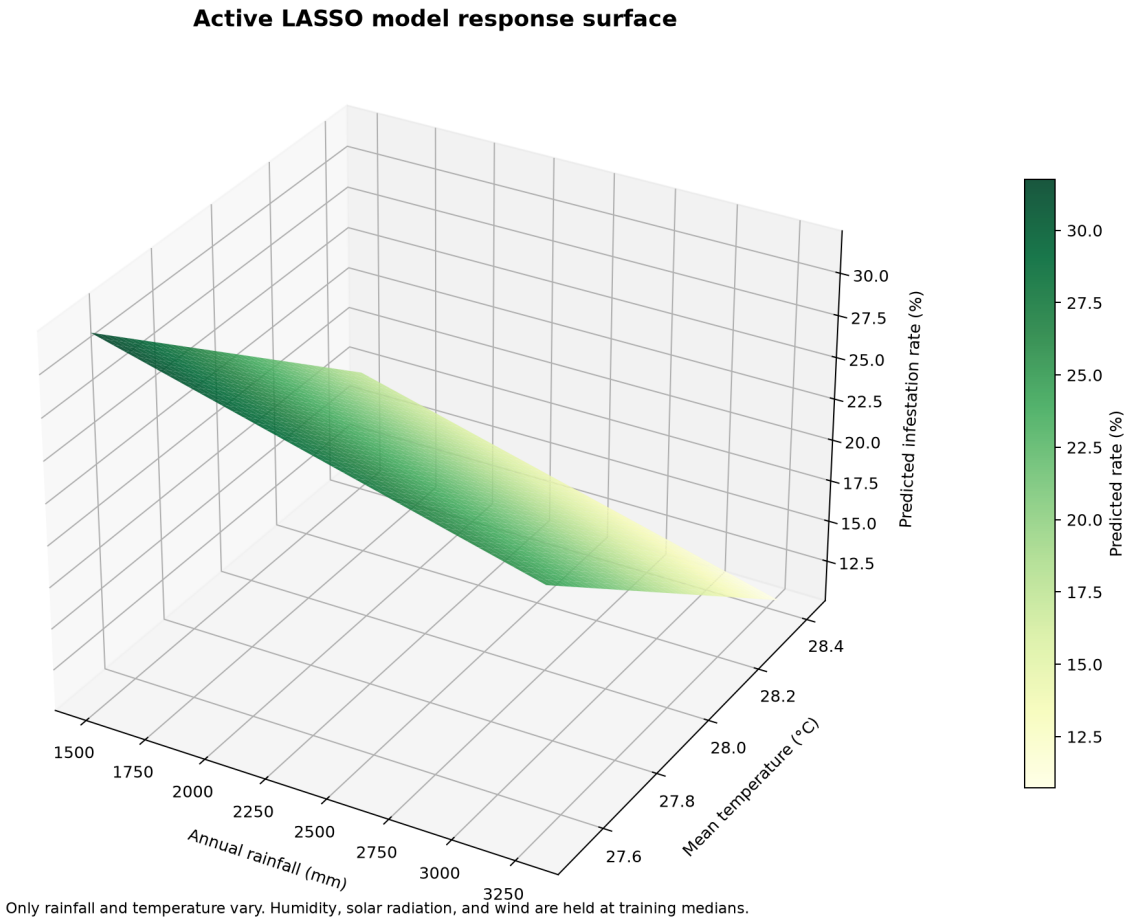
Data provenance: the infestation target is mixed



The 100 training rows combine three documented target constructions. These counts show why comparable future collection should use one defined measurement and aggregation method.

3D SYSTEM VISUALIZATION

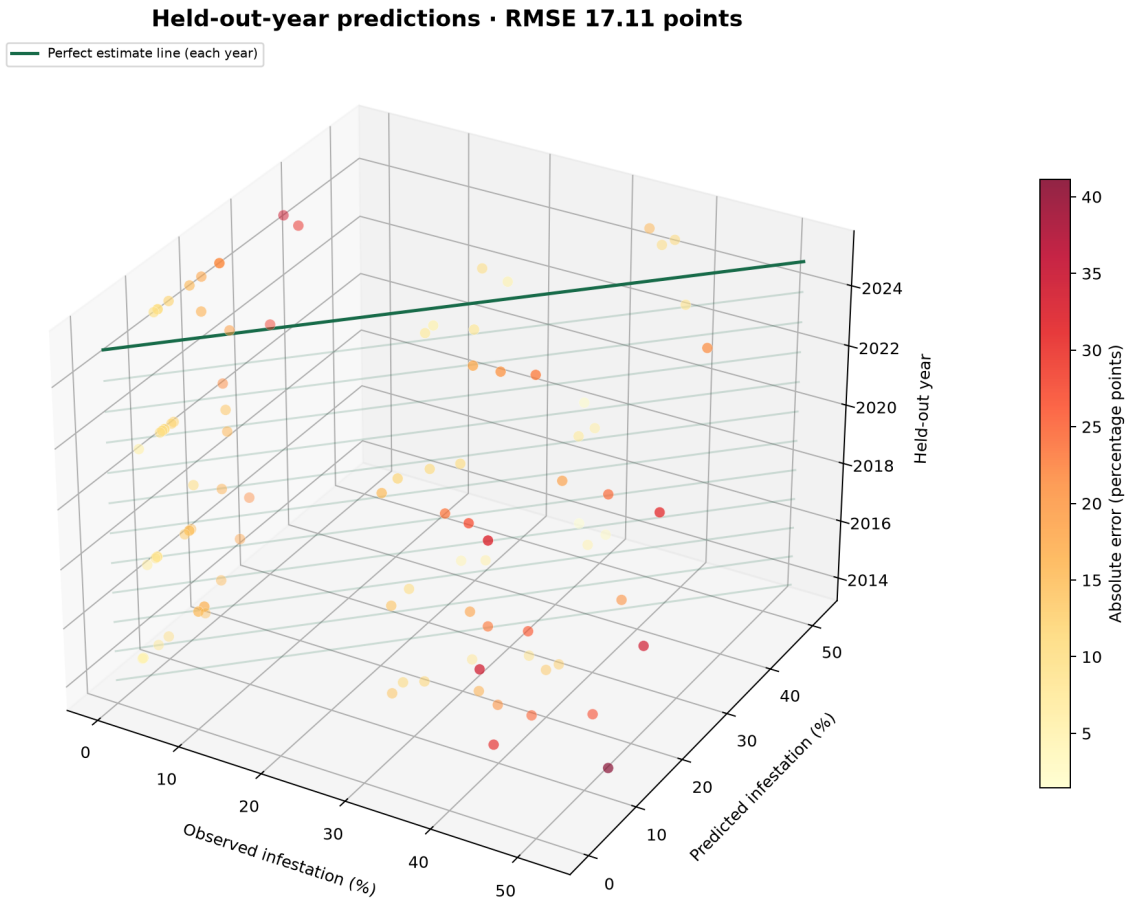
Model view: active LASSO response surface



The surface is conditional on the other three weather variables being fixed at their training-set medians. It visualizes model behavior inside the observed weather range; it is not a map of Palawan or causal evidence.

3D SYSTEM VISUALIZATION

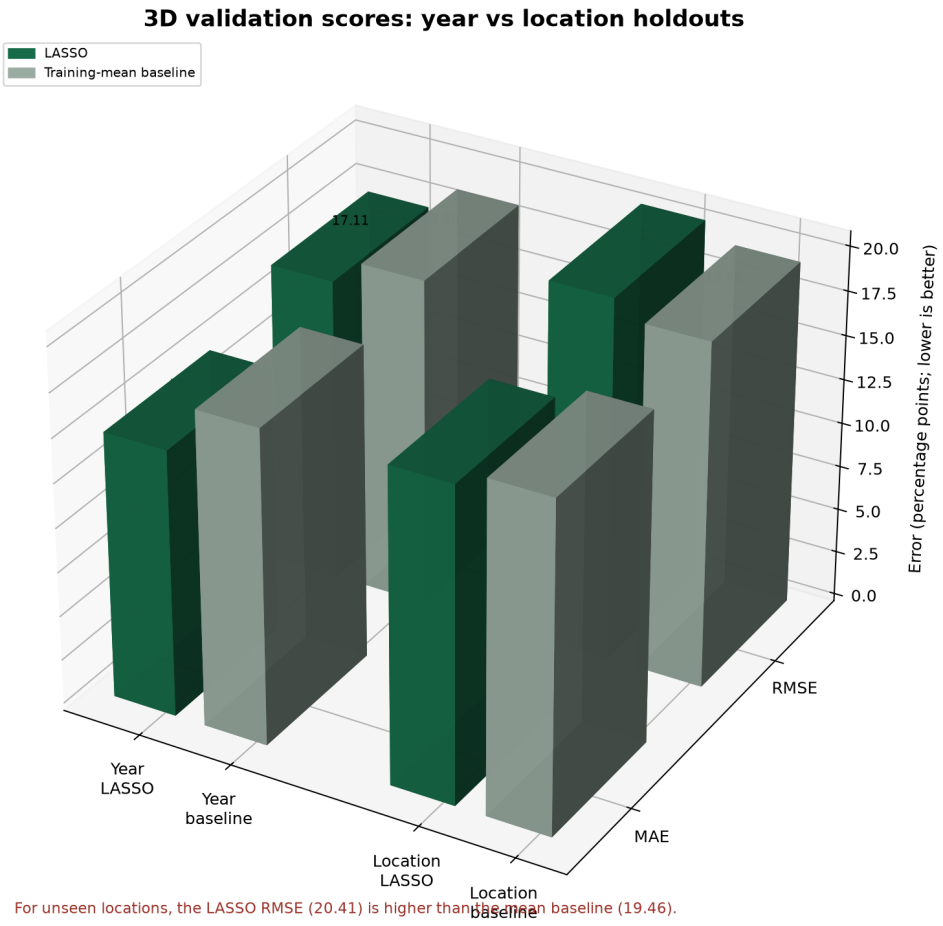
Validation view: estimates for years left out of training



Points near the equal-value line have smaller errors. Color shows absolute error in percentage points. RMSE summarizes the full held-out set and is not the error for every individual forecast.

3D SYSTEM VISUALIZATION

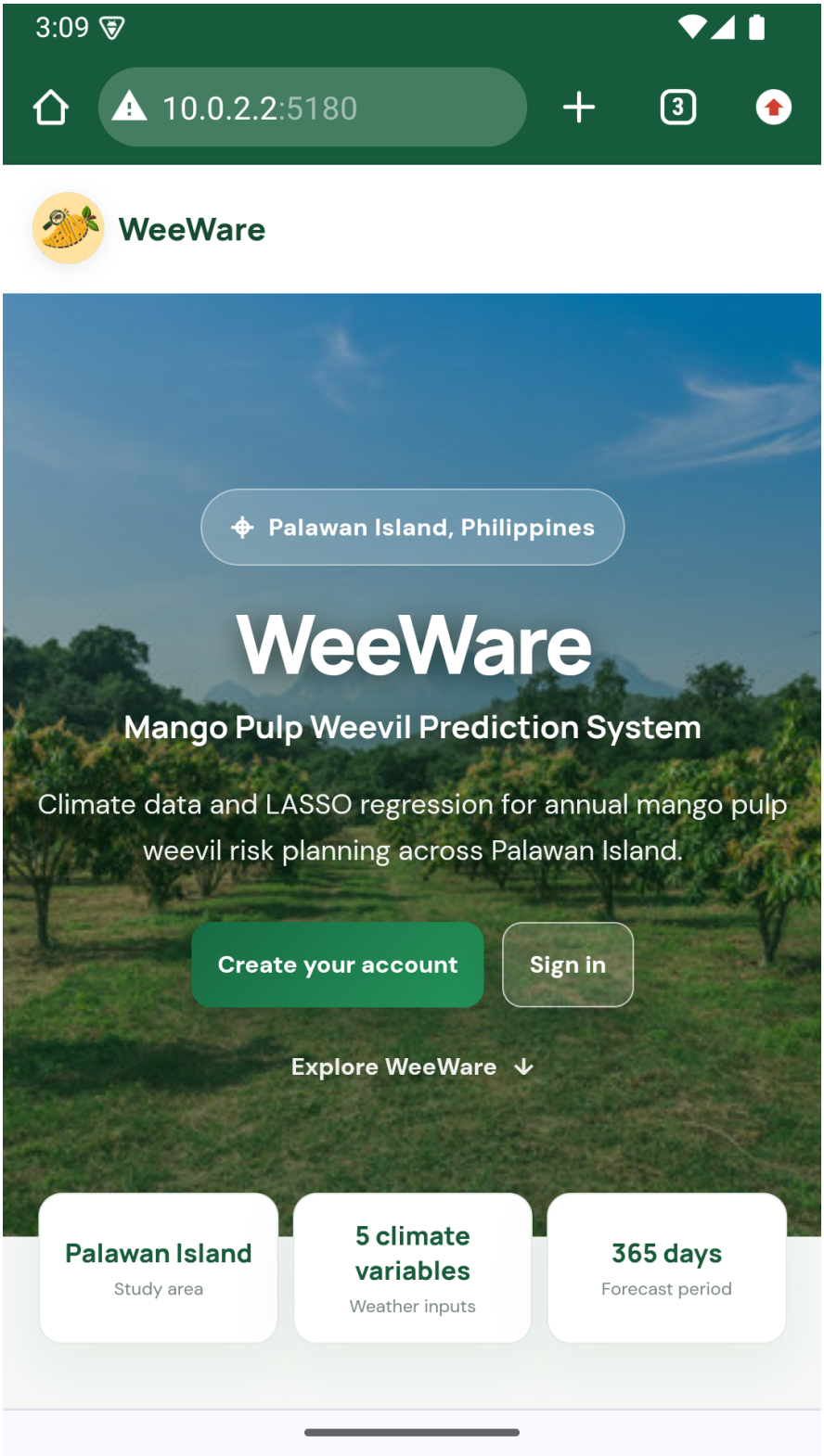
Validation comparison: held-out years and locations



Lower is better. Year-held-out LASSO RMSE is 17.11 points. For location-held-out rows, LASSO RMSE is 20.41 versus 19.46 for the simple mean baseline, highlighting the main concern for predictions at places not represented in training.

Home and about

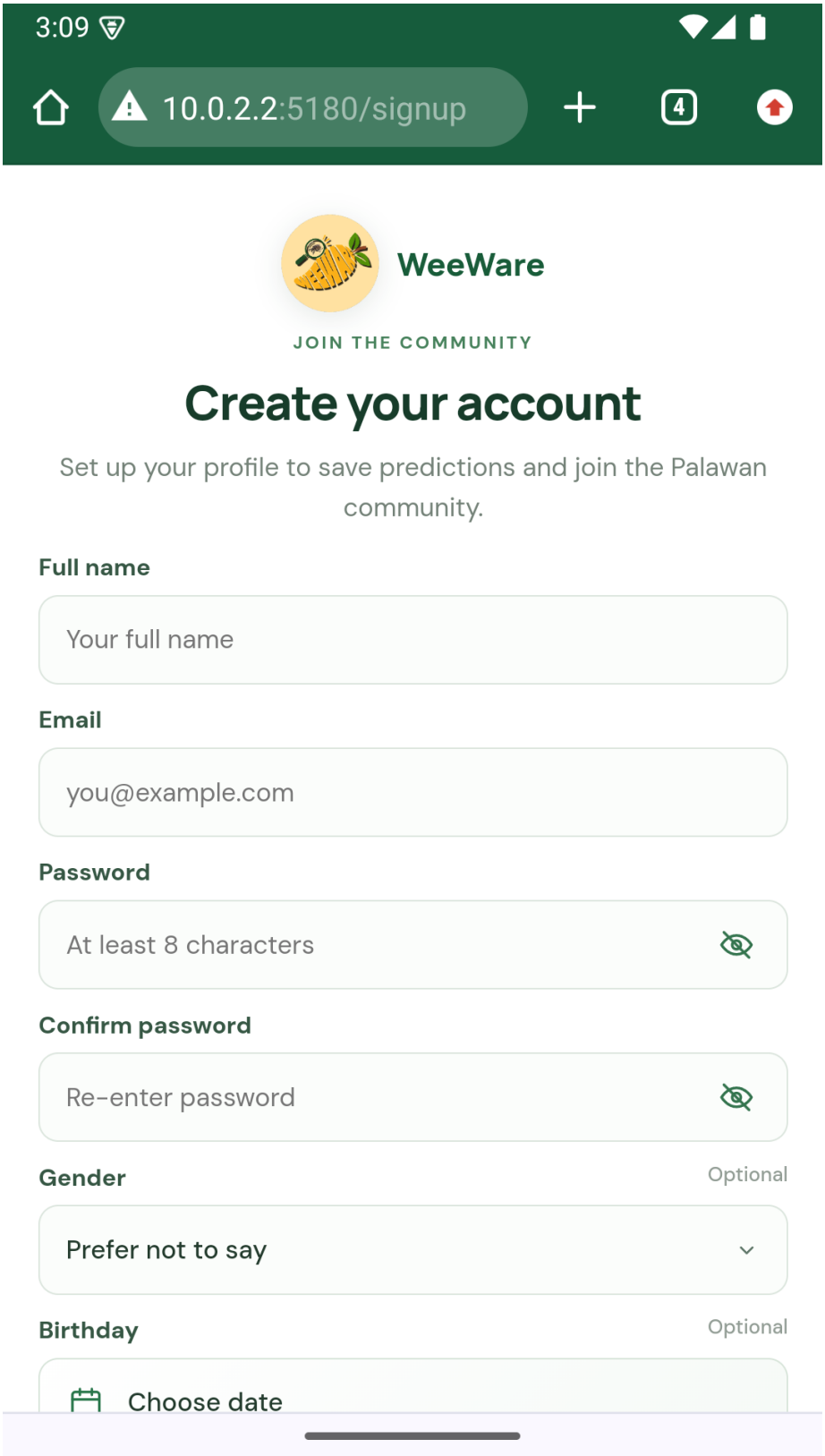
These direct captures come from the local Android virtual device at 1080 x 1920. Examples in account settings, history, forum, chat, and dashboards use synthetic demonstration data.



Public introduction, Palawan-wide scope, and links to create an account or sign in.
File: screenshots/01-home.png

MOBILE SCREEN CAPTURE

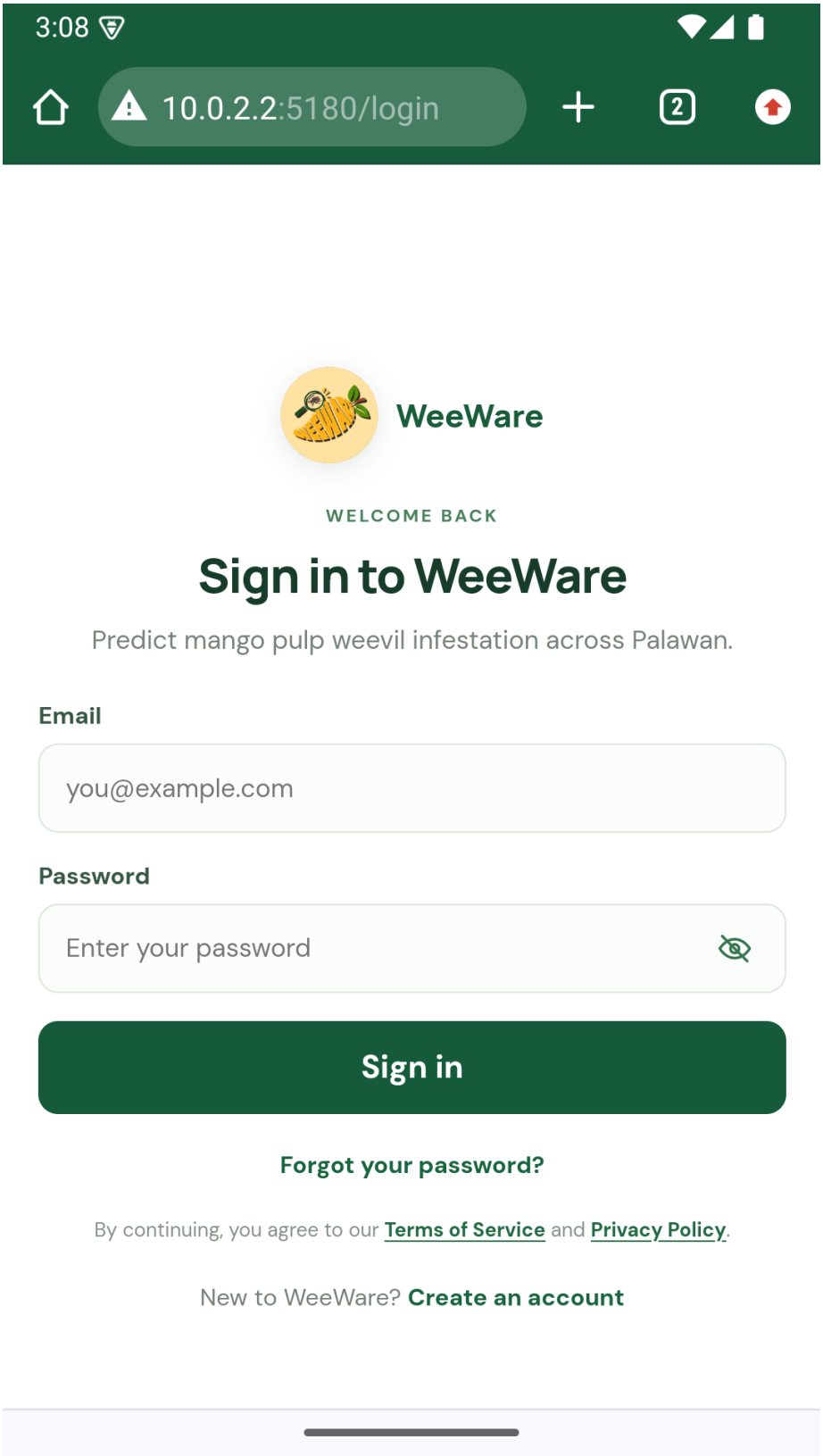
Create an account



Mobile signup fields, optional profile information, and the email verification entry flow.
File: screenshots/02-create-account.png

MOBILE SCREEN CAPTURE

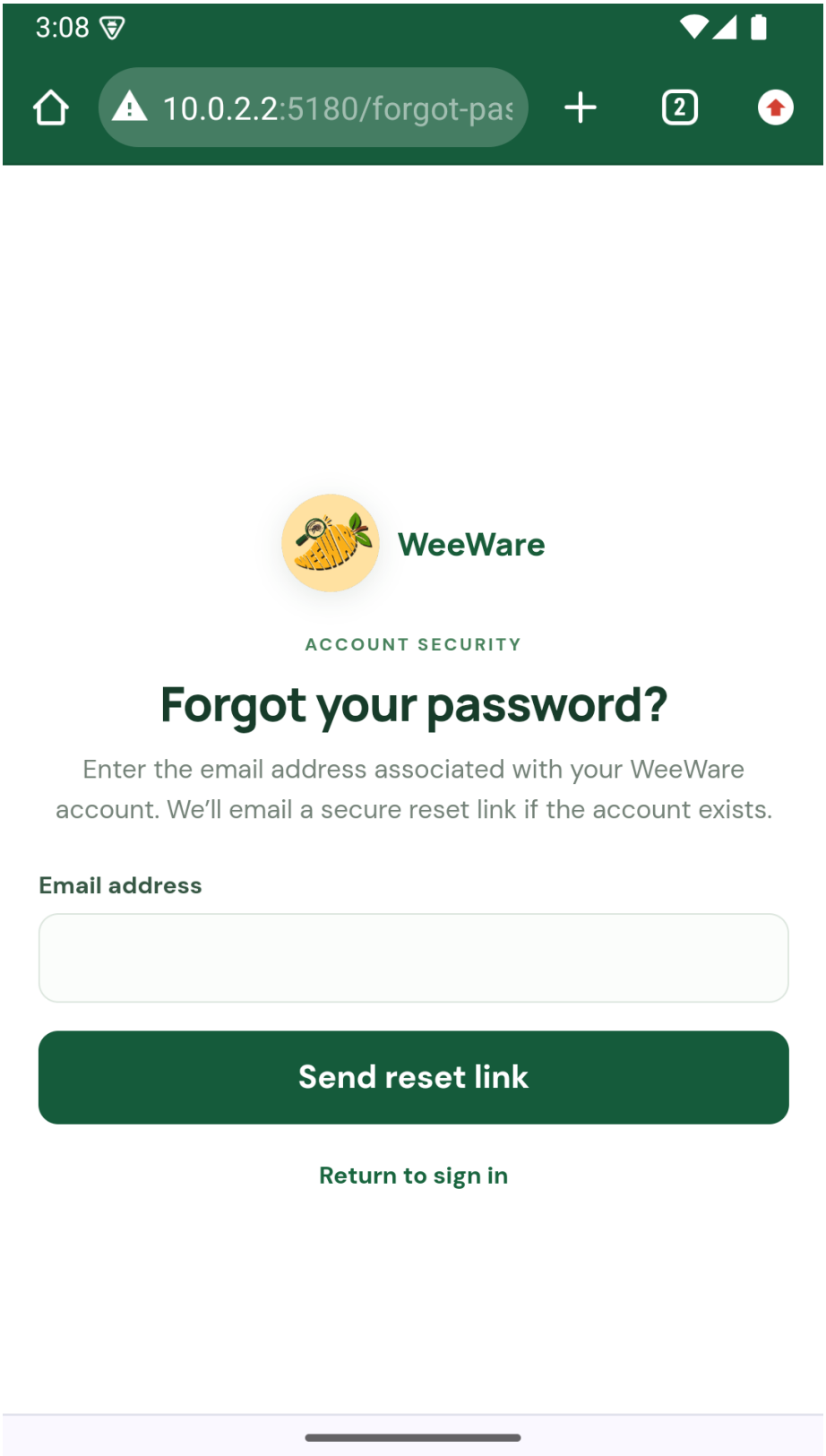
Sign in



Email/password form and password visibility control.
File: screenshots/03-sign-in.png

MOBILE SCREEN CAPTURE

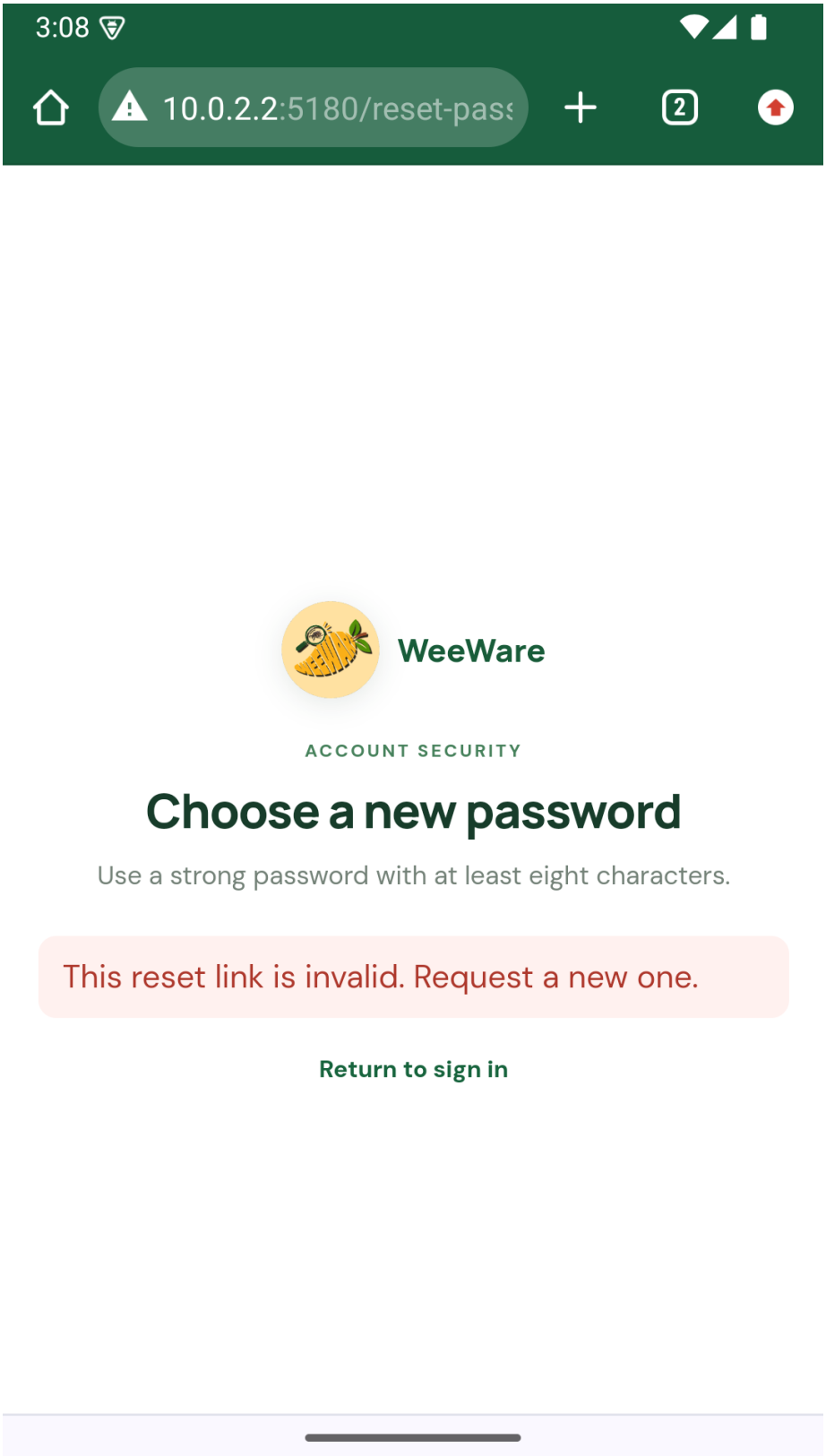
Forgot password



Secure password recovery request form.
File: screenshots/04-forgot-password.png

MOBILE SCREEN CAPTURE

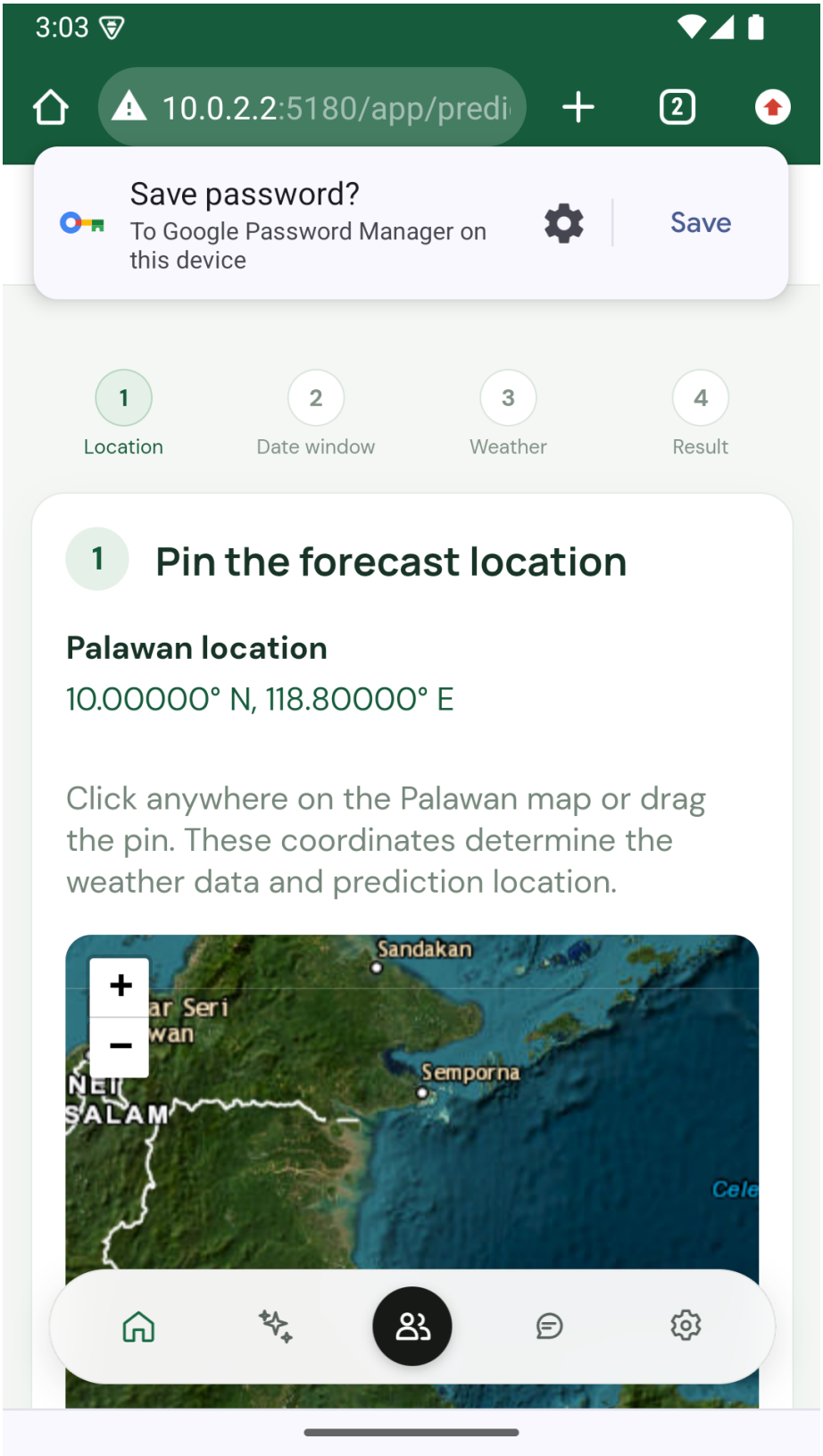
Reset password



Reset page for a missing/invalid token; a valid emailed link displays the new-password form.
File: screenshots/05-reset-password.png

MOBILE SCREEN CAPTURE

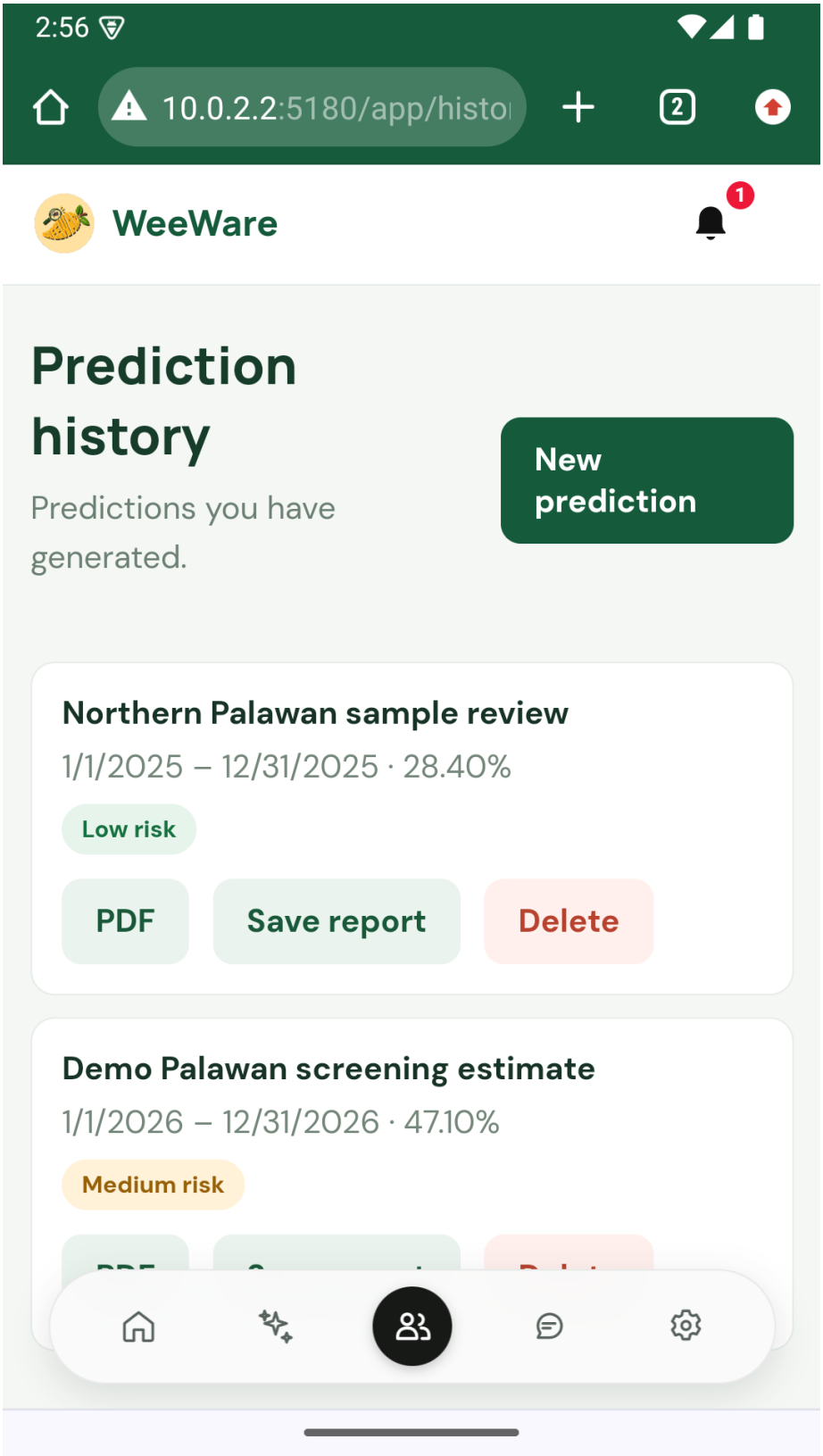
Prediction: select a location



First wizard step, coordinates, and interactive Palawan map.
File: screenshots/06-prediction-location.png

MOBILE SCREEN CAPTURE

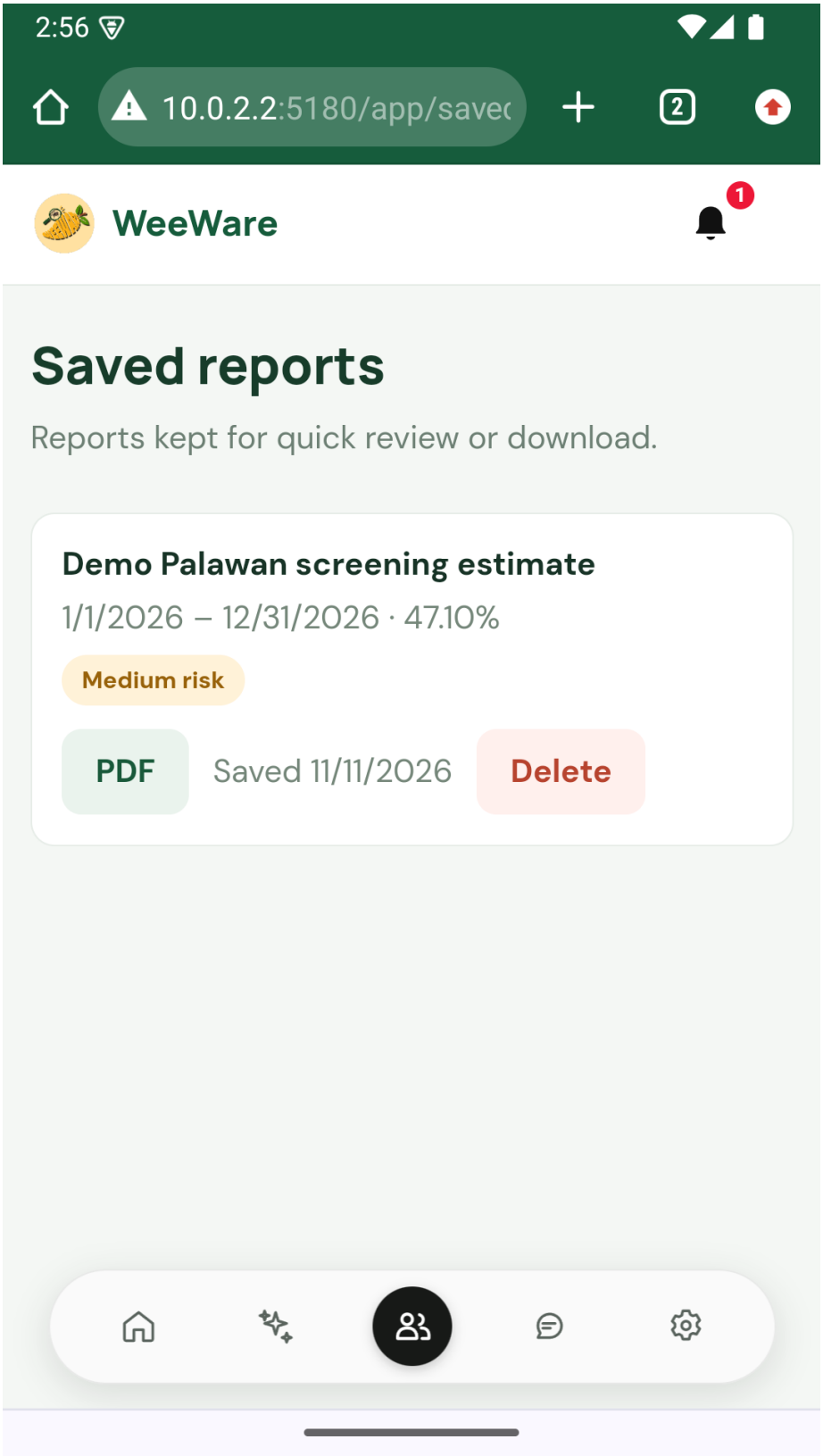
Prediction history



Saved forecast history and per-result actions.
File: screenshots/07-prediction-history.png

MOBILE SCREEN CAPTURE

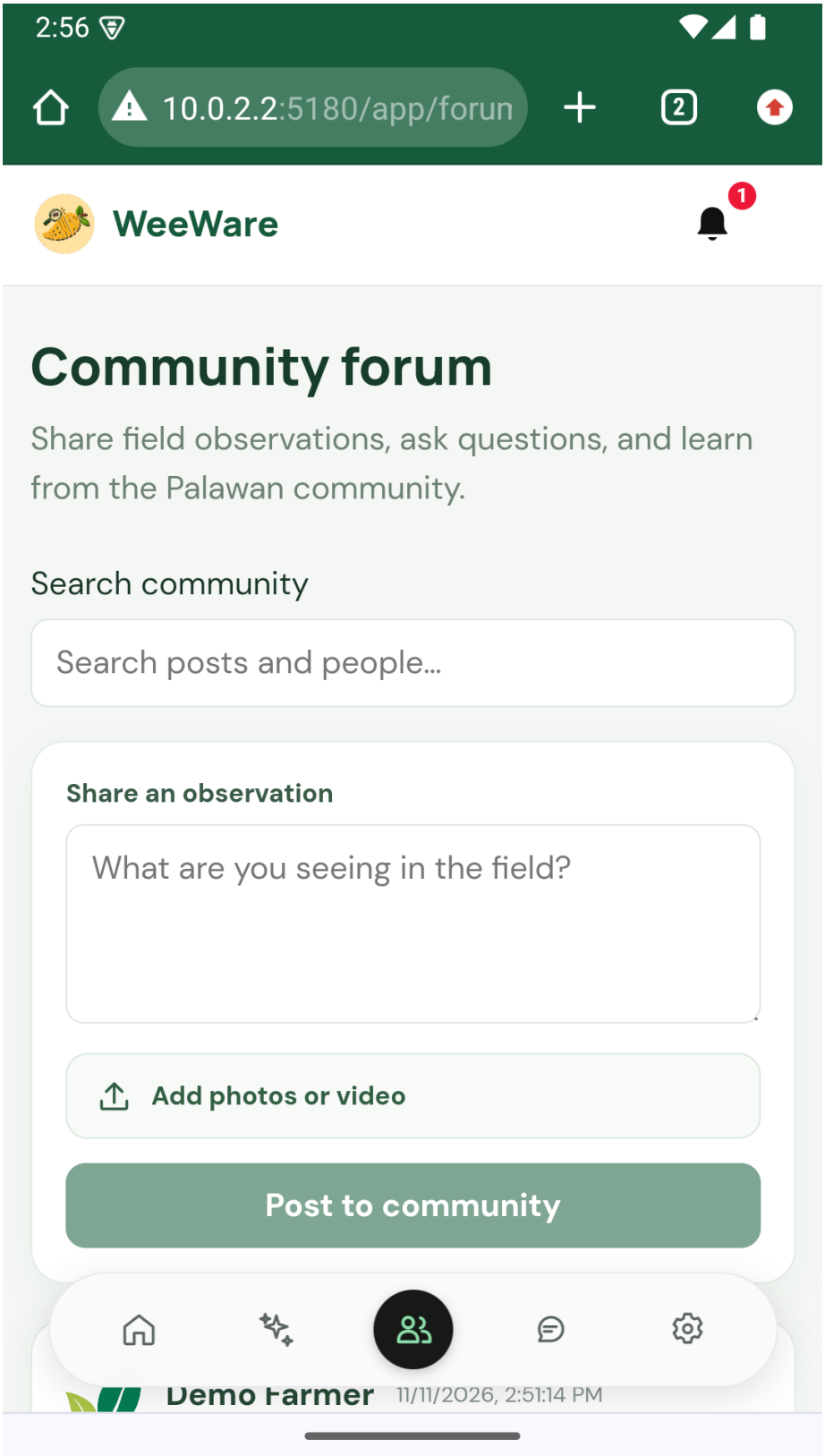
Saved reports



Forecasts retained for quick reference and download.
File: screenshots/08-saved-reports.png

MOBILE SCREEN CAPTURE

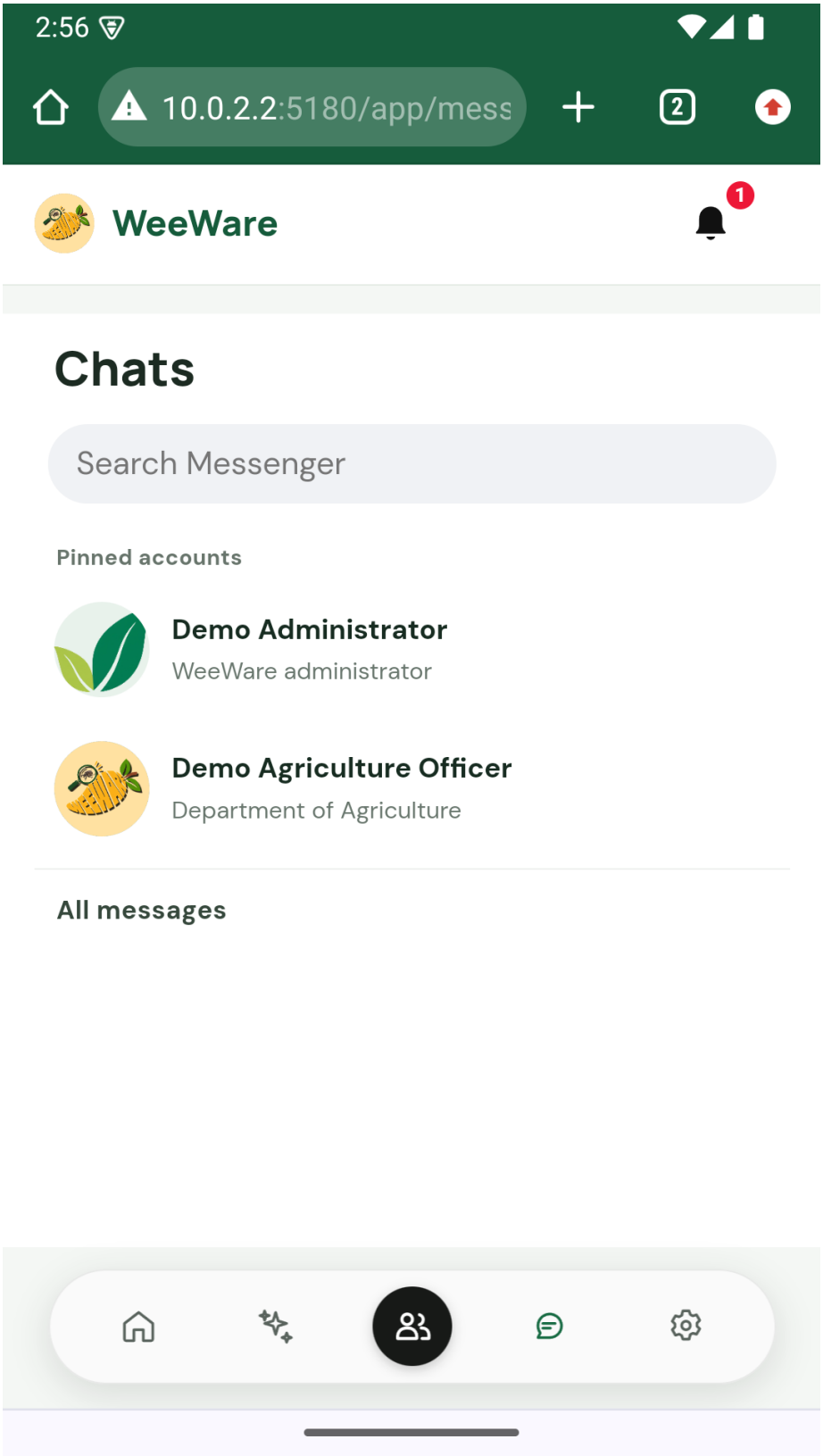
Community forum



Search, compose a post, add media, and browse discussions.
File: screenshots/09-community-forum.png

MOBILE SCREEN CAPTURE

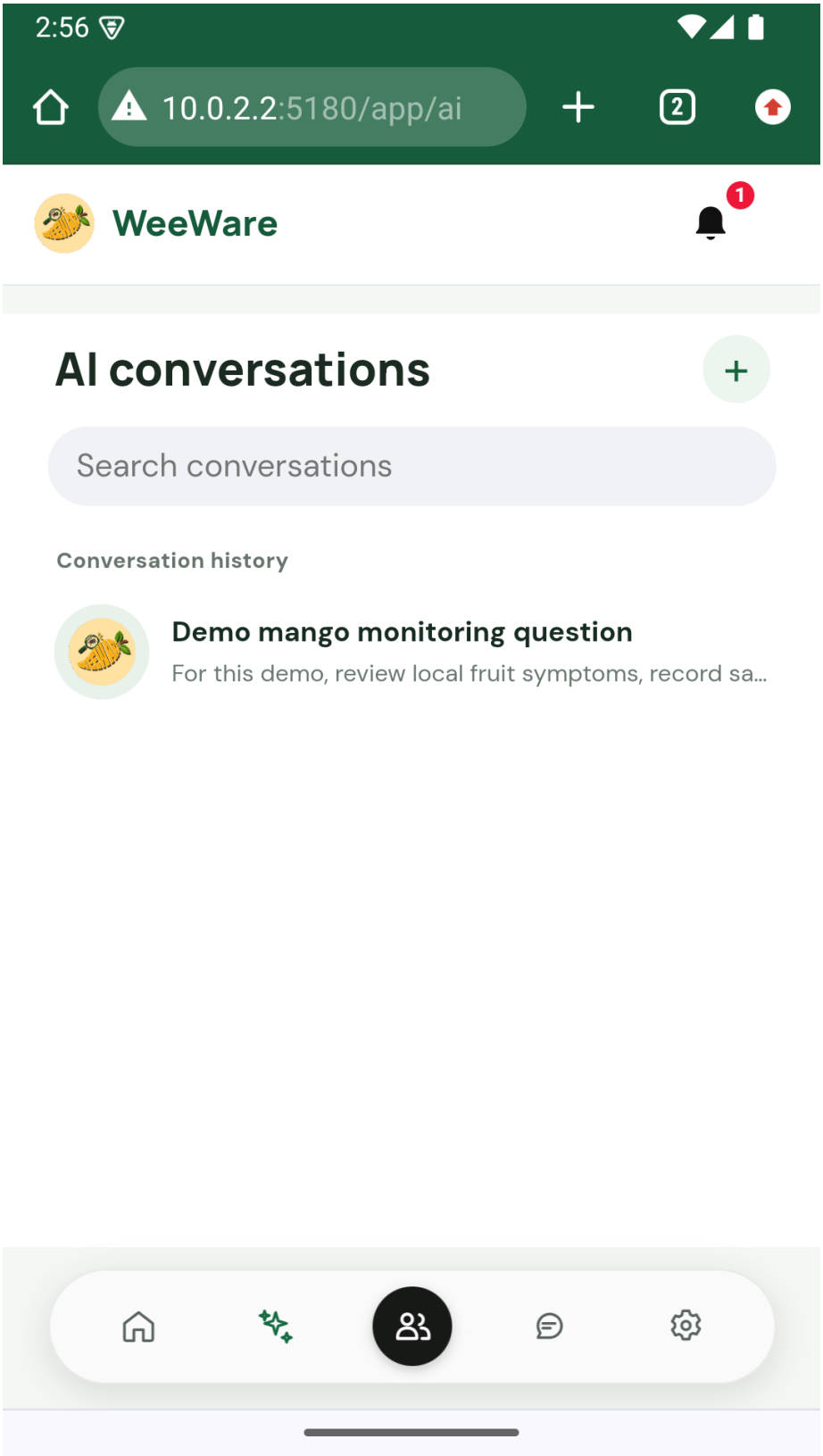
Messages inbox



Direct conversations and role-pinned accounts.
File: screenshots/10-messages.png

MOBILE SCREEN CAPTURE

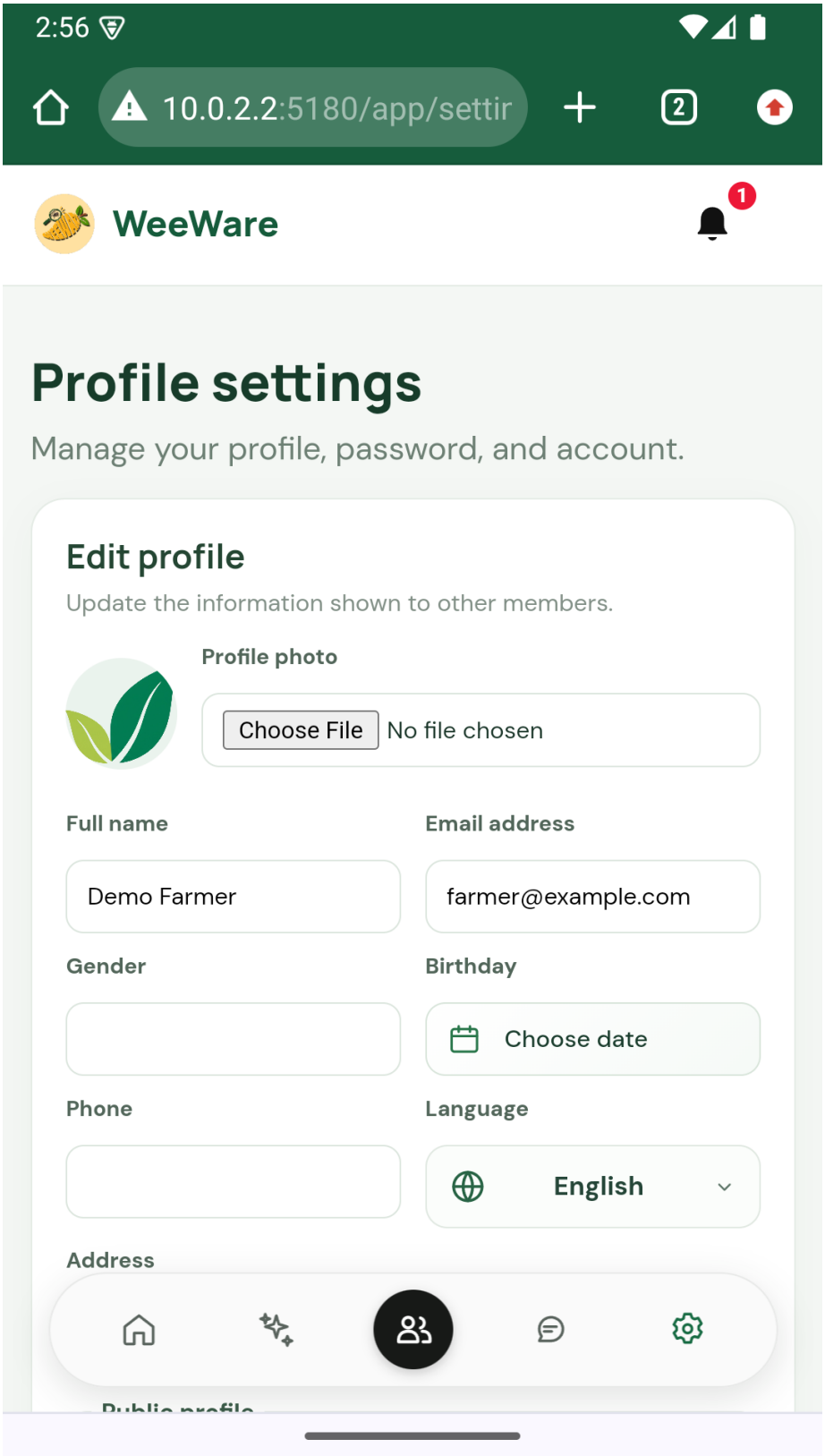
AI assistant



Conversation list and assistant message interface.
File: screenshots/11-ai-assistant.png

MOBILE SCREEN CAPTURE

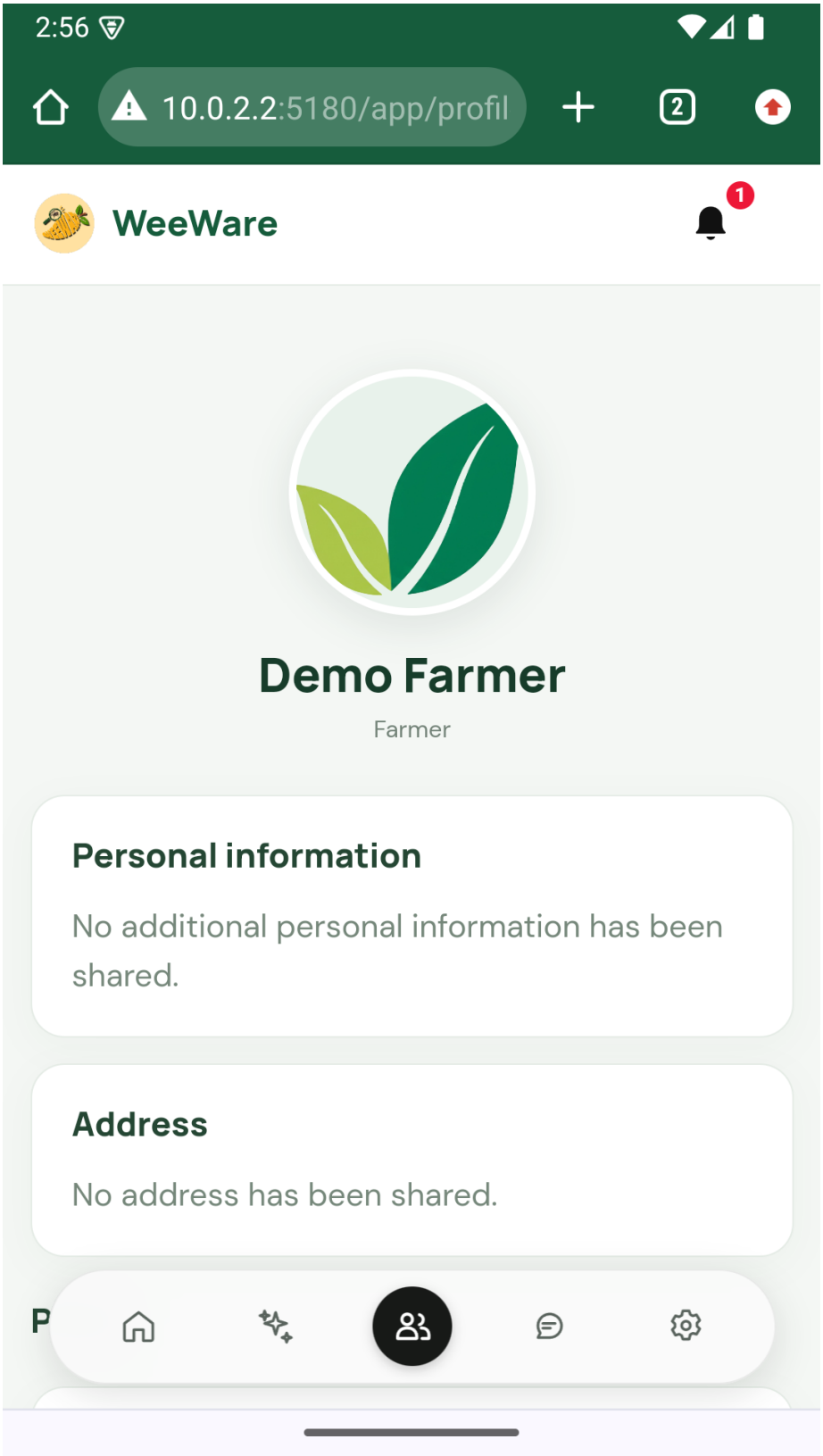
Profile settings



Profile, photo, language, and public visibility fields.
File: screenshots/12-settings.png

MOBILE SCREEN CAPTURE

Member profile



Public profile view and available member actions.
File: screenshots/13-user-profile.png

MOBILE SCREEN CAPTURE

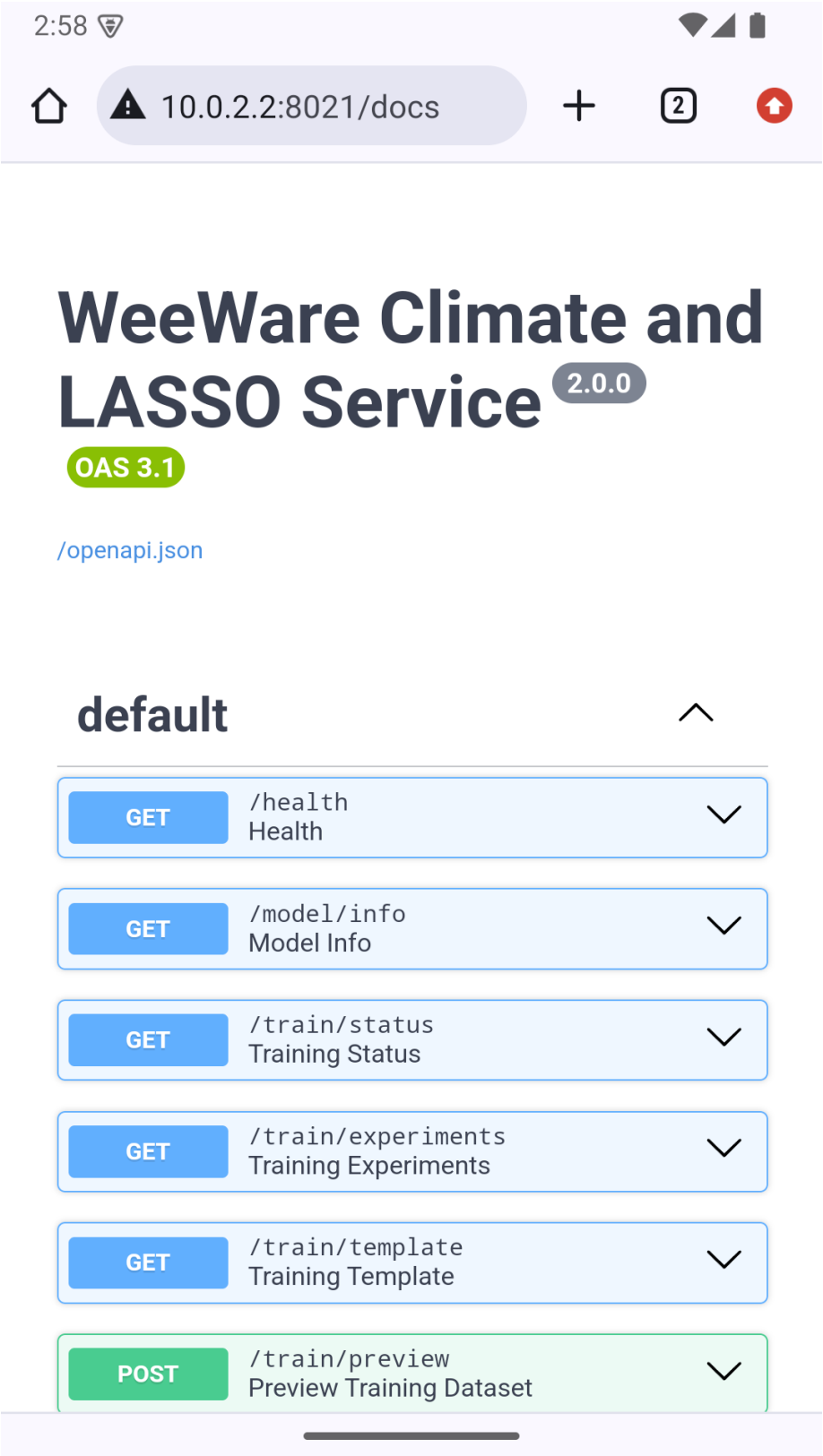
Express API health



The health endpoint returns service status and timestamp as JSON.
File: screenshots/14-api-health.png

MOBILE SCREEN CAPTURE

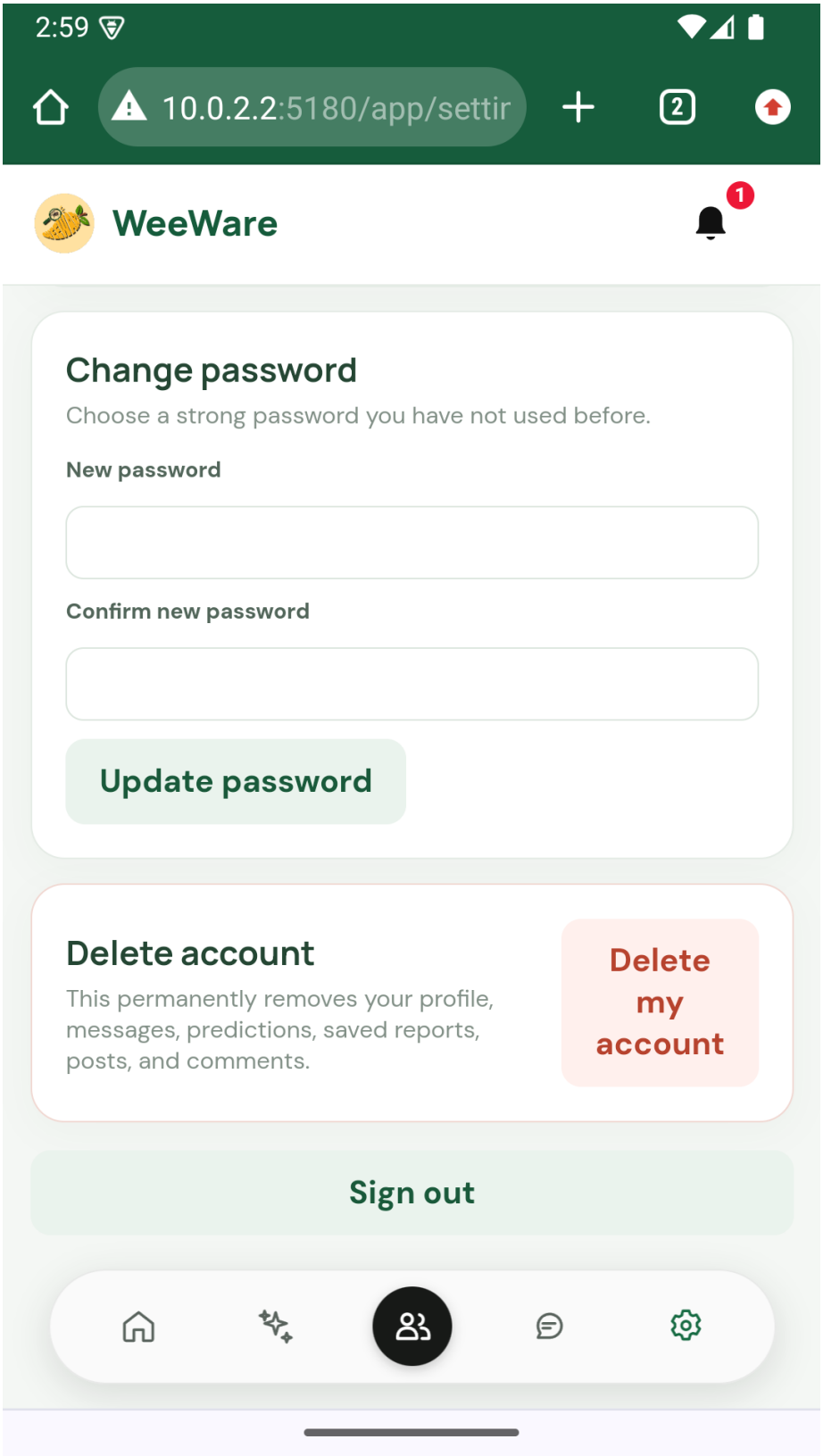
Climate/model API reference



FastAPI OpenAPI page lists the climate, prediction, and researcher operations.
File: screenshots/15-model-api-reference.png

MOBILE SCREEN CAPTURE

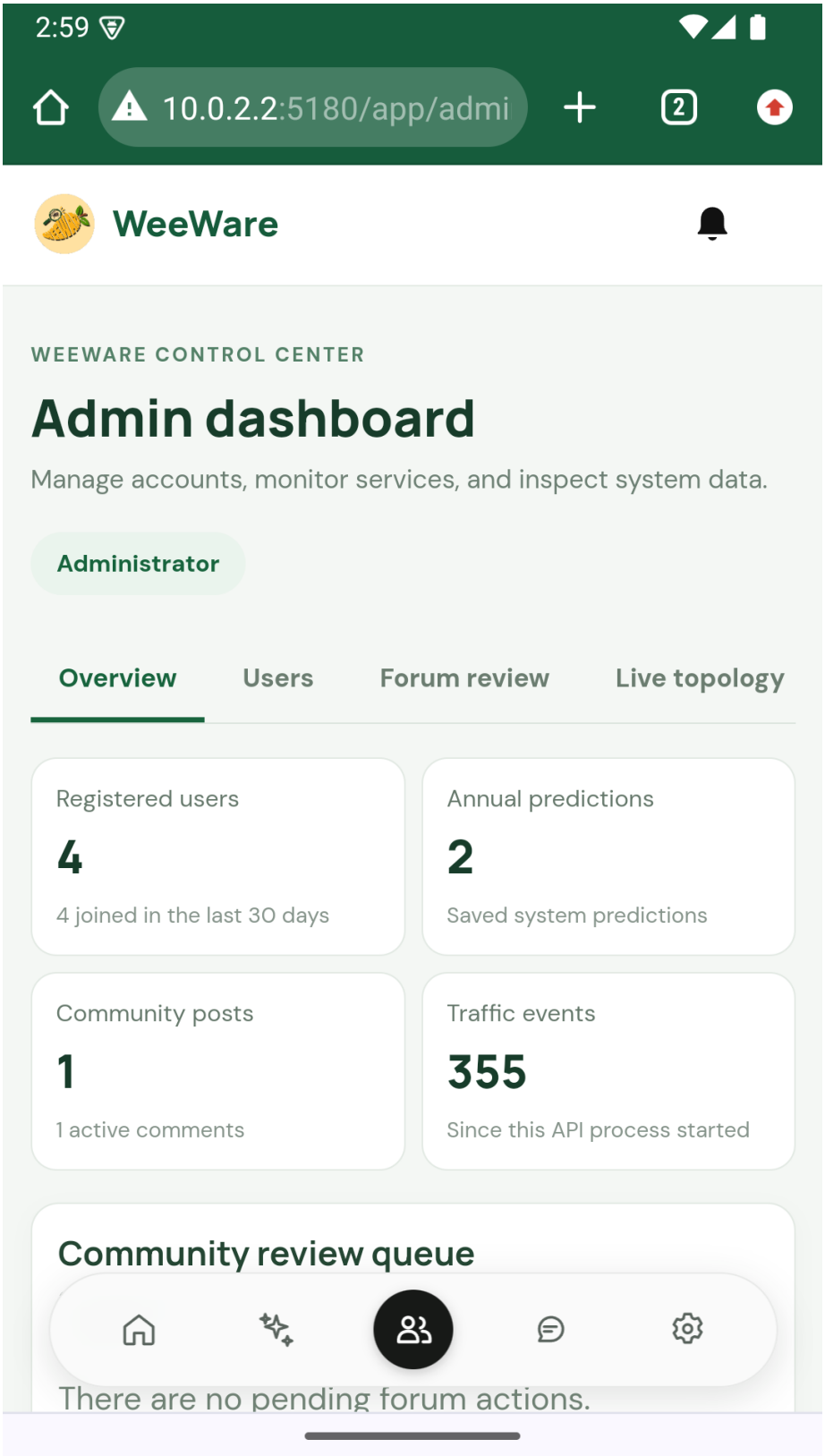
Account controls



Password change, account deletion, and sign-out actions.
File: screenshots/16-account-controls.png

MOBILE SCREEN CAPTURE

Admin dashboard



System overview, account/content controls, and monitoring tabs.
File: screenshots/17-admin-dashboard.png

MOBILE SCREEN CAPTURE

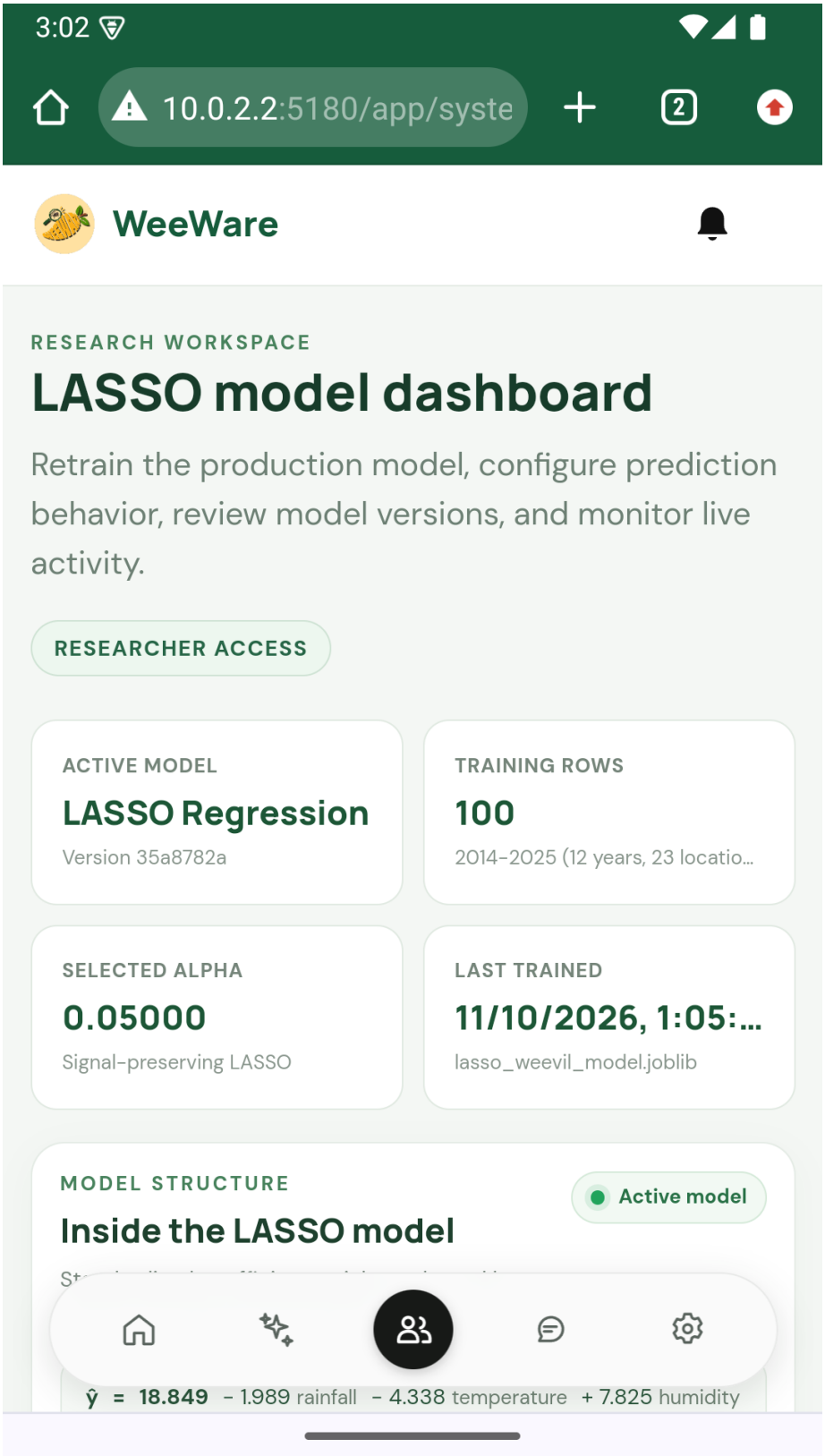
Designated group chat



Admin/DA/Researcher shared conversation on mobile.
File: screenshots/18-admin-group-chat.png

MOBILE SCREEN CAPTURE

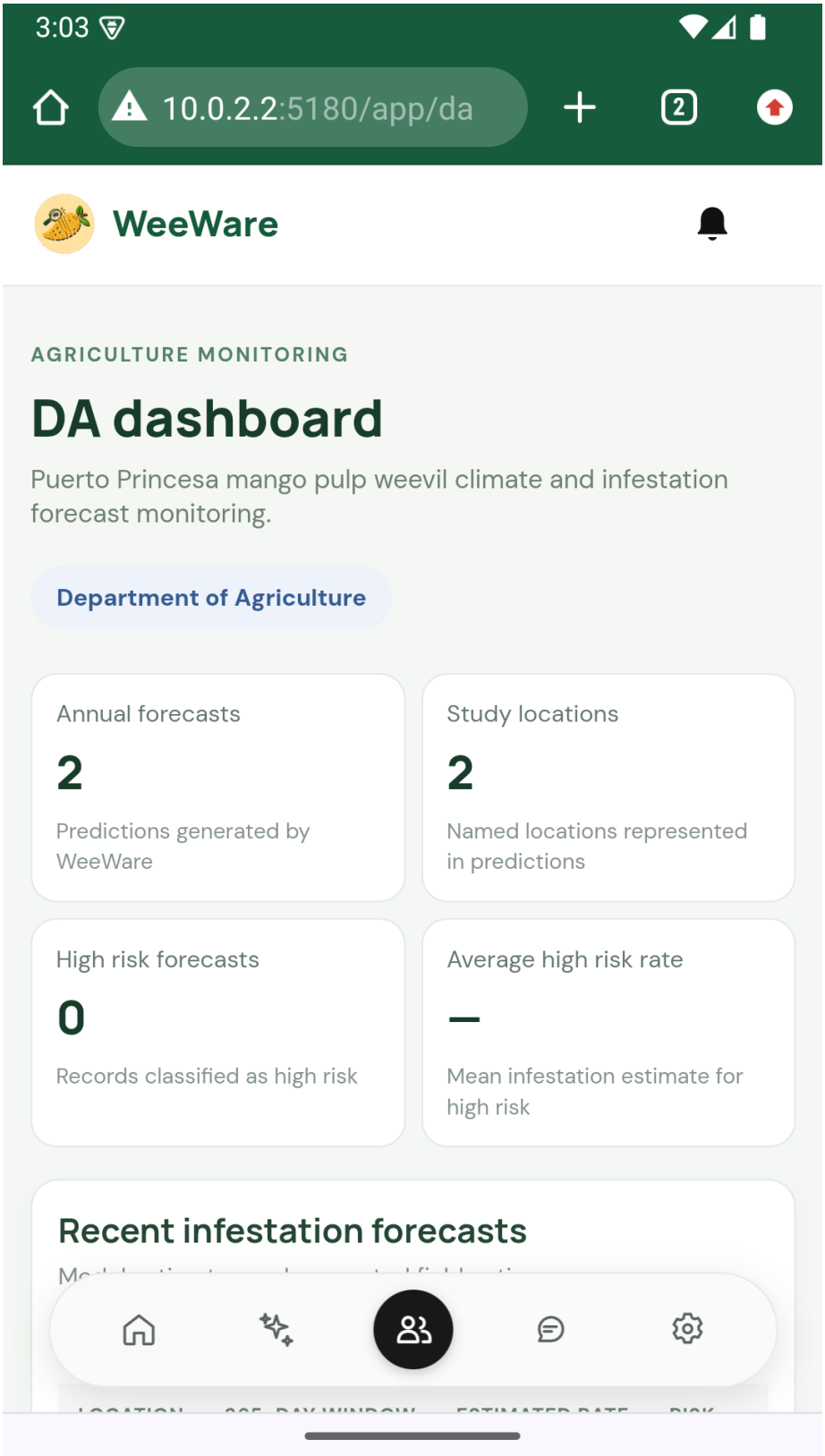
Researcher dashboard



Active Palawan LASSO version, training evidence, and retraining workspace.
File: screenshots/19-researcher-dashboard.png

MOBILE SCREEN CAPTURE

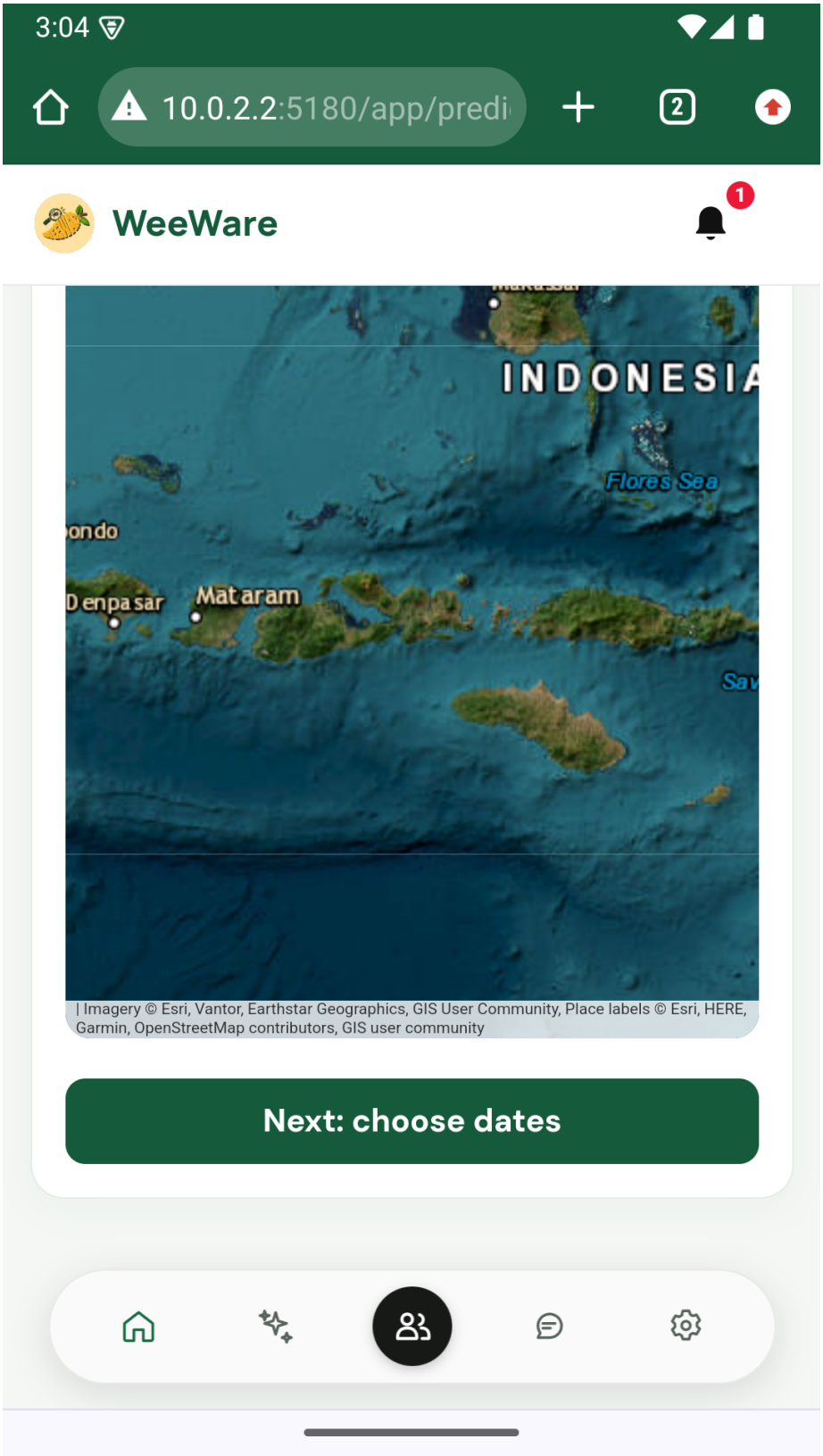
DA dashboard



Forecast and location summaries for agriculture monitoring.
File: screenshots/20-da-dashboard.png

MOBILE SCREEN CAPTURE

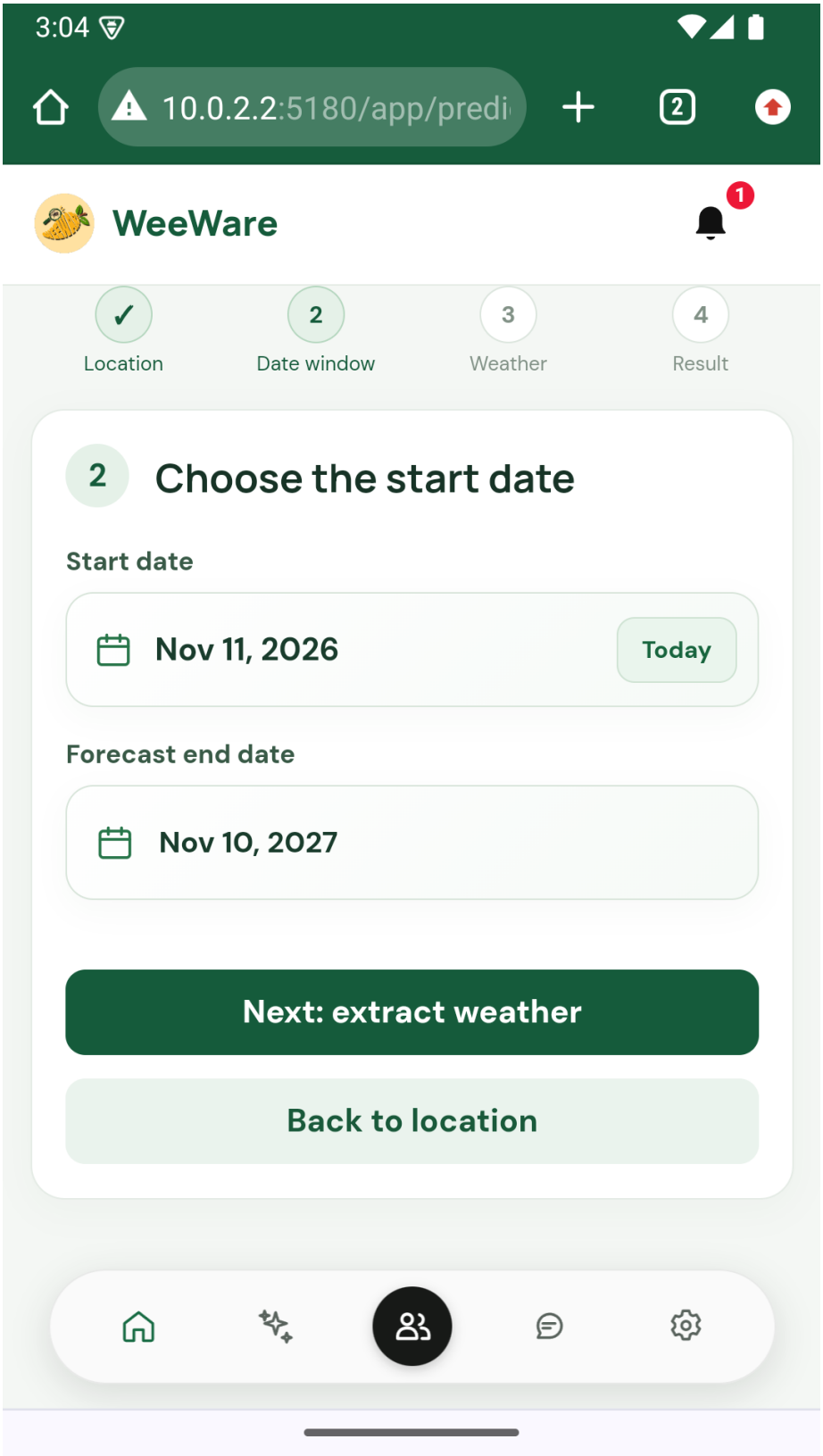
Location step control



Map-step button advances the wizard after choosing the pin.
File: screenshots/21-prediction-location-action.png

MOBILE SCREEN CAPTURE

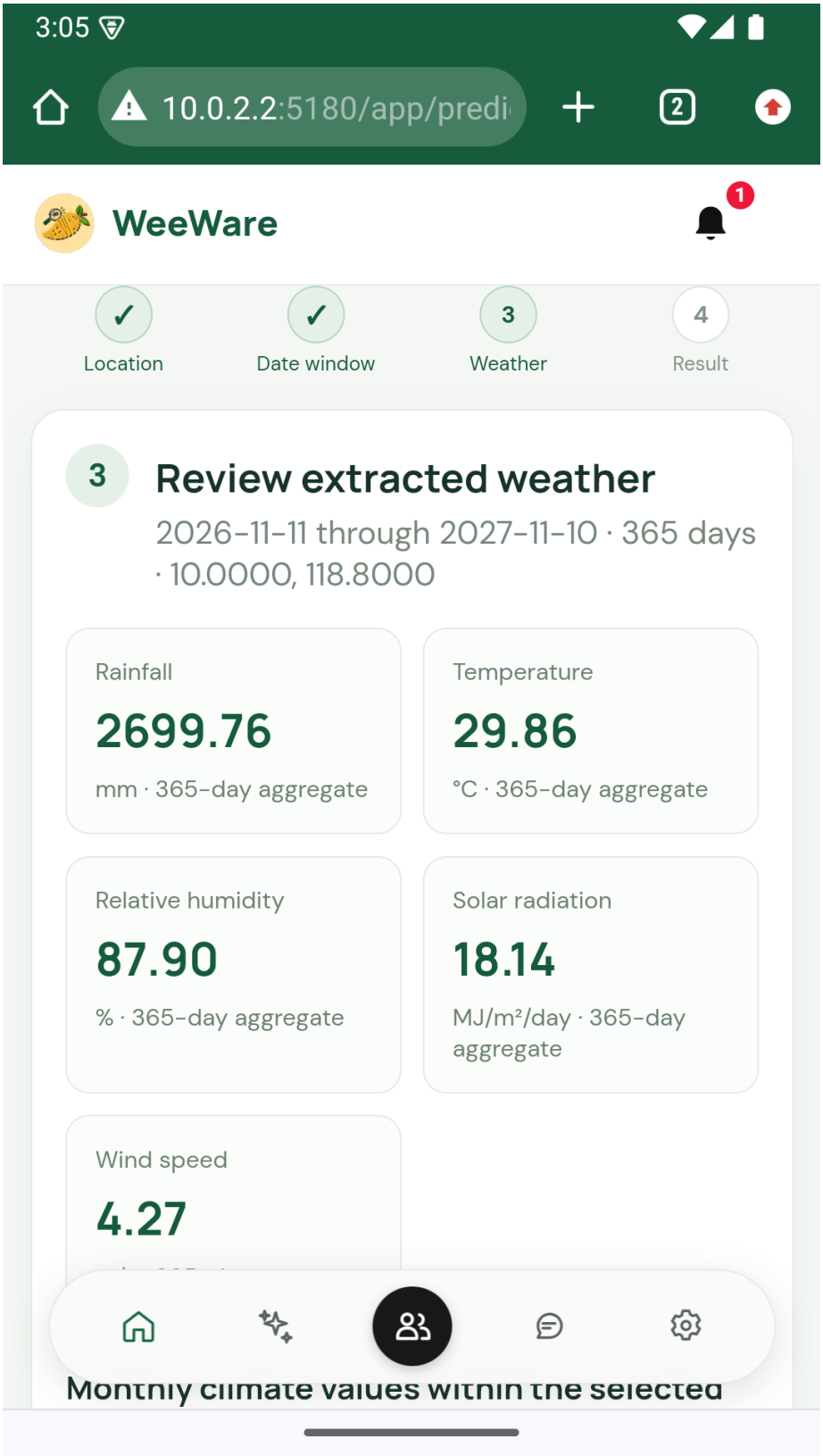
Date window control



A start date and a forecast end date exactly 365 calendar days apart.
File: screenshots/22-prediction-date-window.png

MOBILE SCREEN CAPTURE

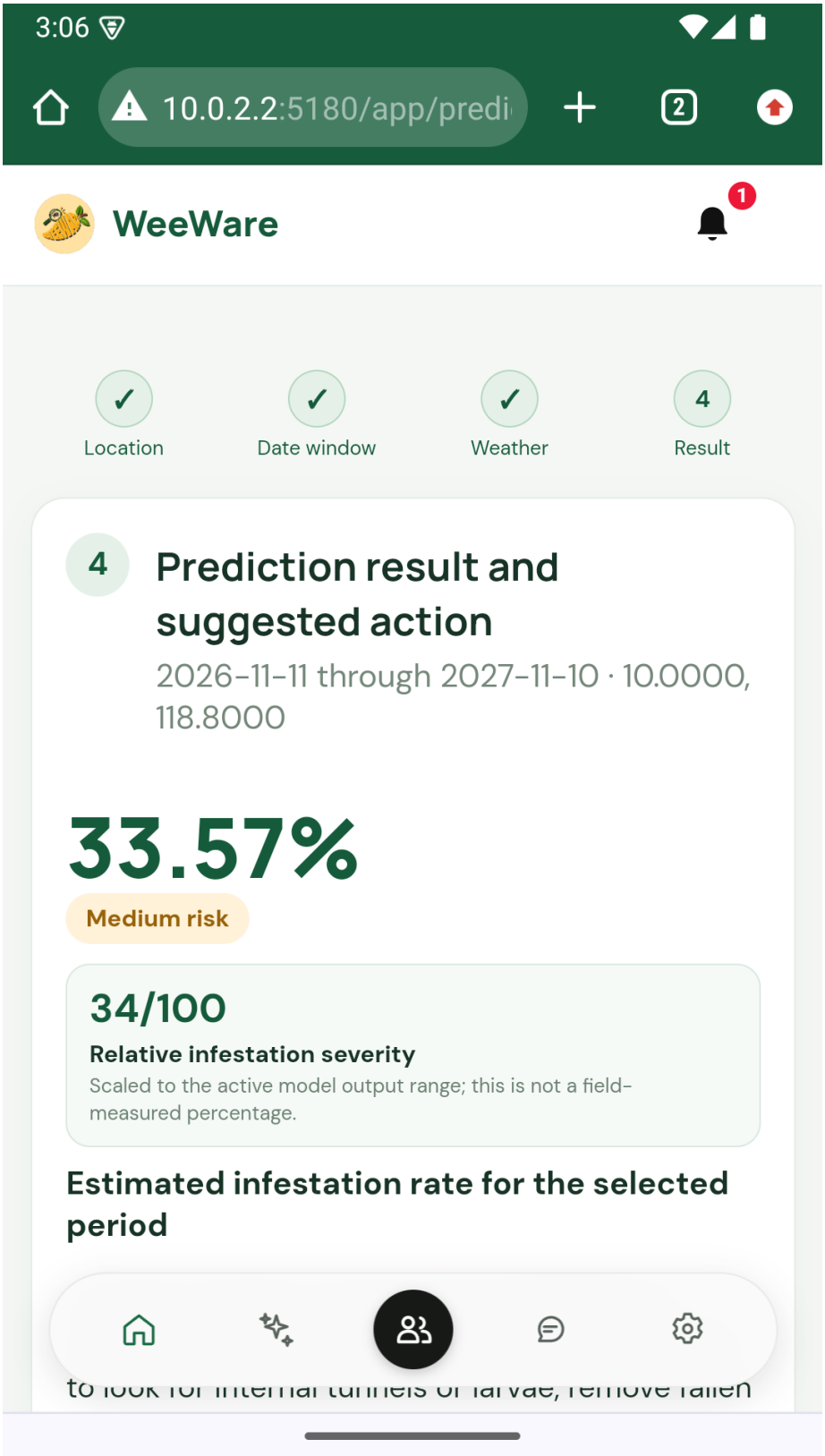
Weather review



Annual totals/averages for five variables and monthly summaries.
File: screenshots/23-prediction-weather-review.png

MOBILE SCREEN CAPTURE

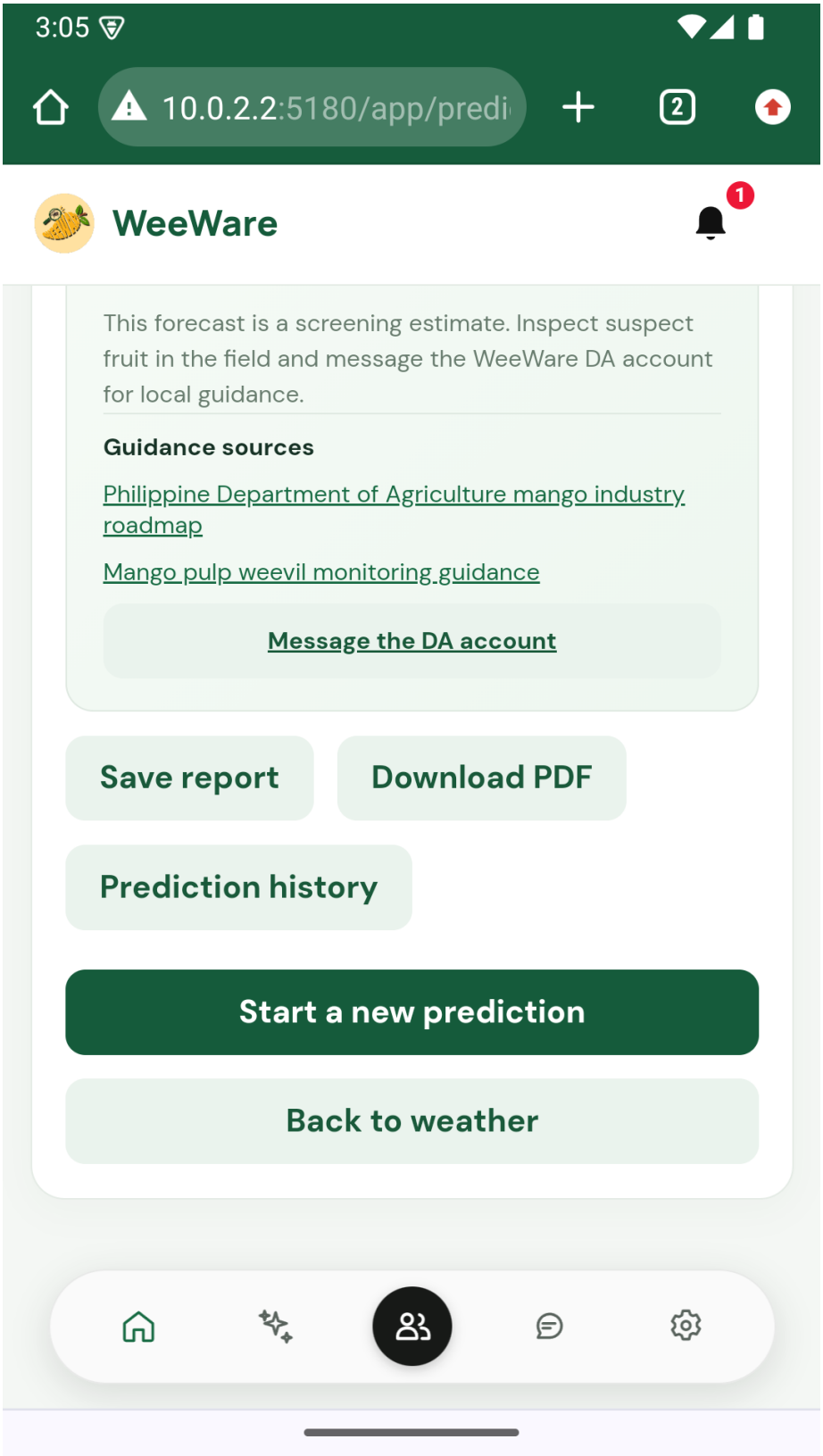
Prediction result



Estimate, risk category, selected period, and interpretation note.
File: screenshots/24-prediction-result.png

MOBILE SCREEN CAPTURE

Result actions



Save report, download PDF, view history, start over, or return to weather.
File: screenshots/25-result-controls.png

MOBILE SCREEN CAPTURE

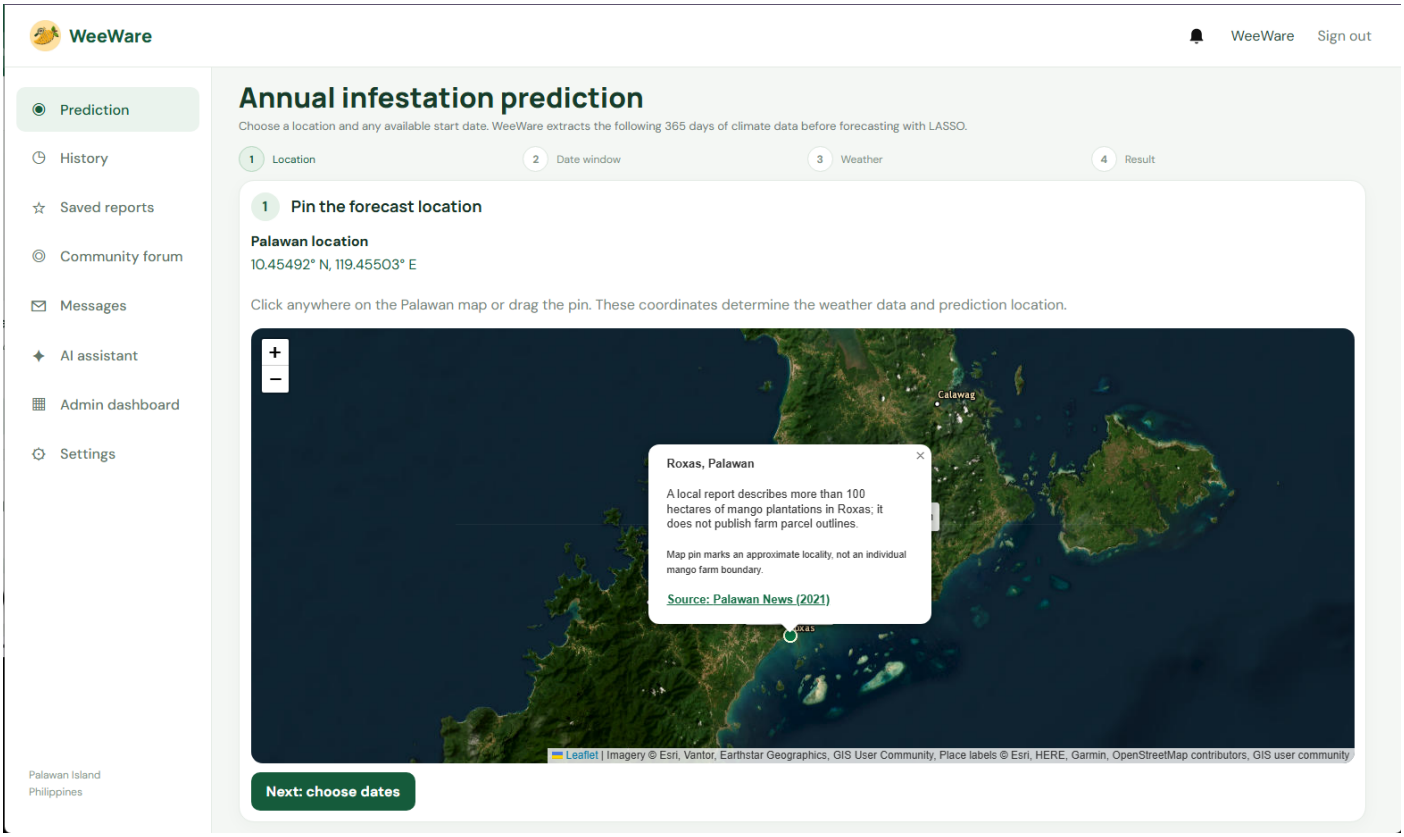
Forum photo viewer



Full-screen media viewer with previous/next, zoom in/out, and close controls.
File: screenshots/26-forum-image-zoom.png

Client screenshot: pinned map location

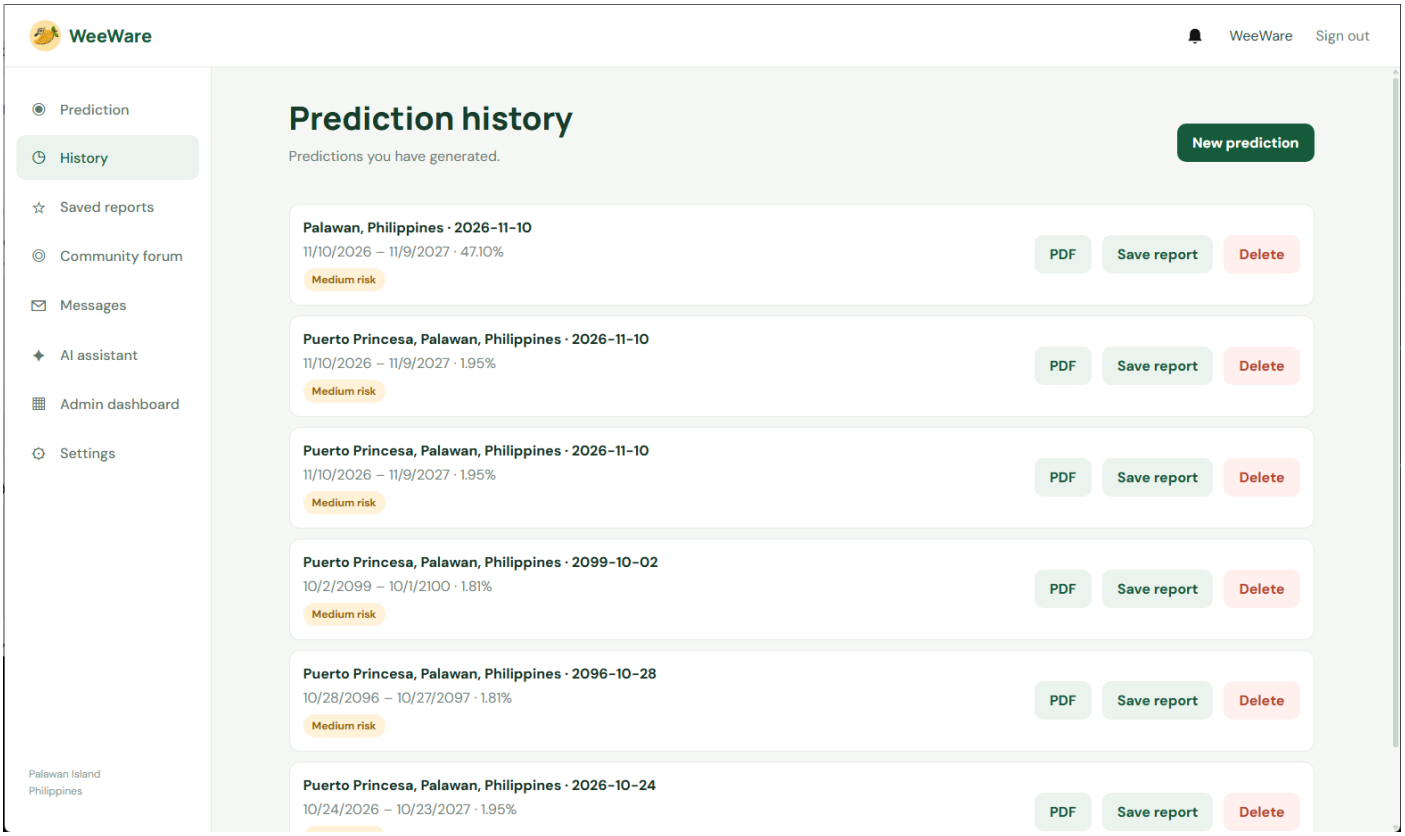
These screenshots were supplied in the sibling `reports/system images` folder. Personal name, email, phone, and address fields are masked in two images before inclusion.



Shows a selected Palawan map point. A pin identifies an approximate locality, not a farm boundary.
File: screenshots/client-supplied/01-location-map.png

CLIENT-PROVIDED SCREENSHOT

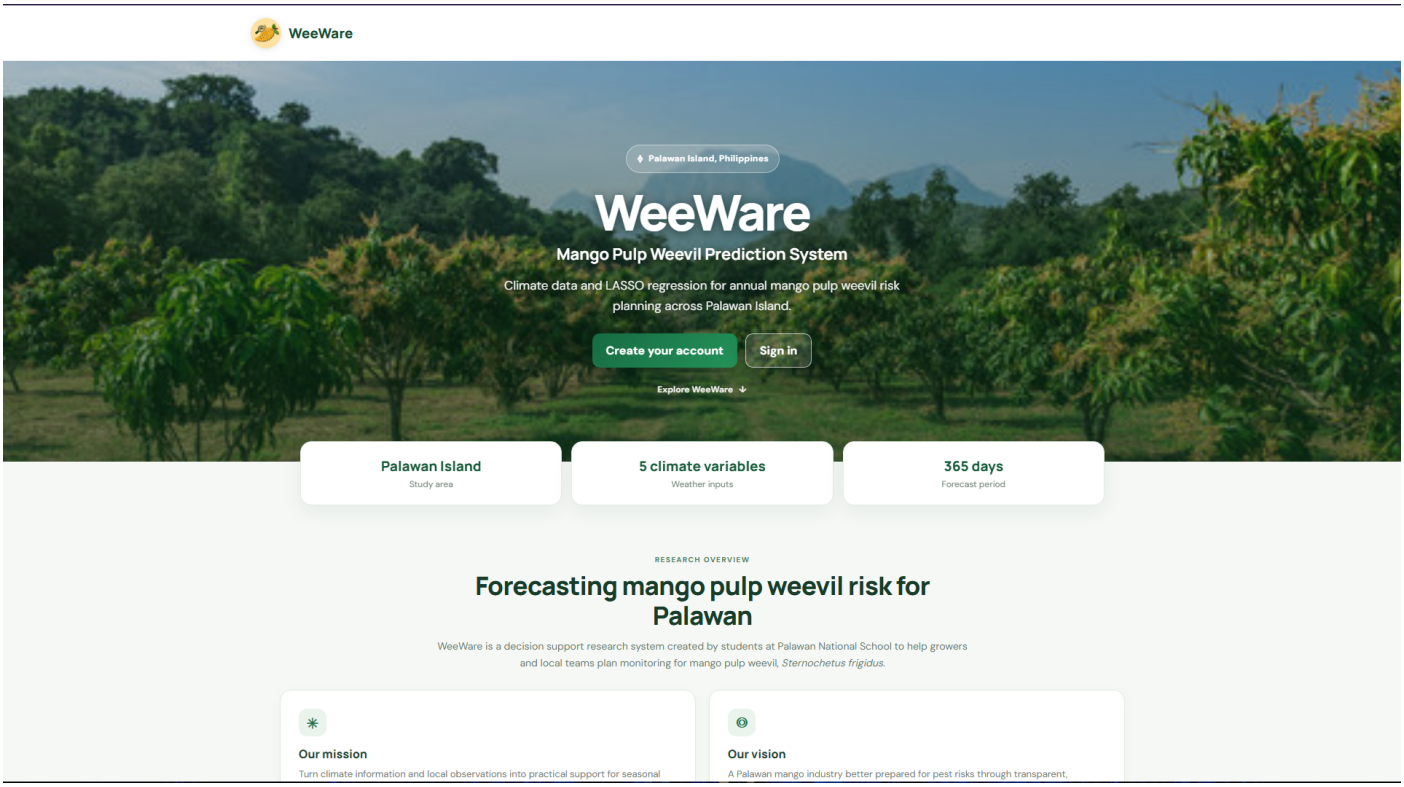
Client screenshot: prediction history



Shows saved forecast history and report controls.
File: screenshots/client-supplied/02-prediction-history.png

CLIENT-PROVIDED SCREENSHOT

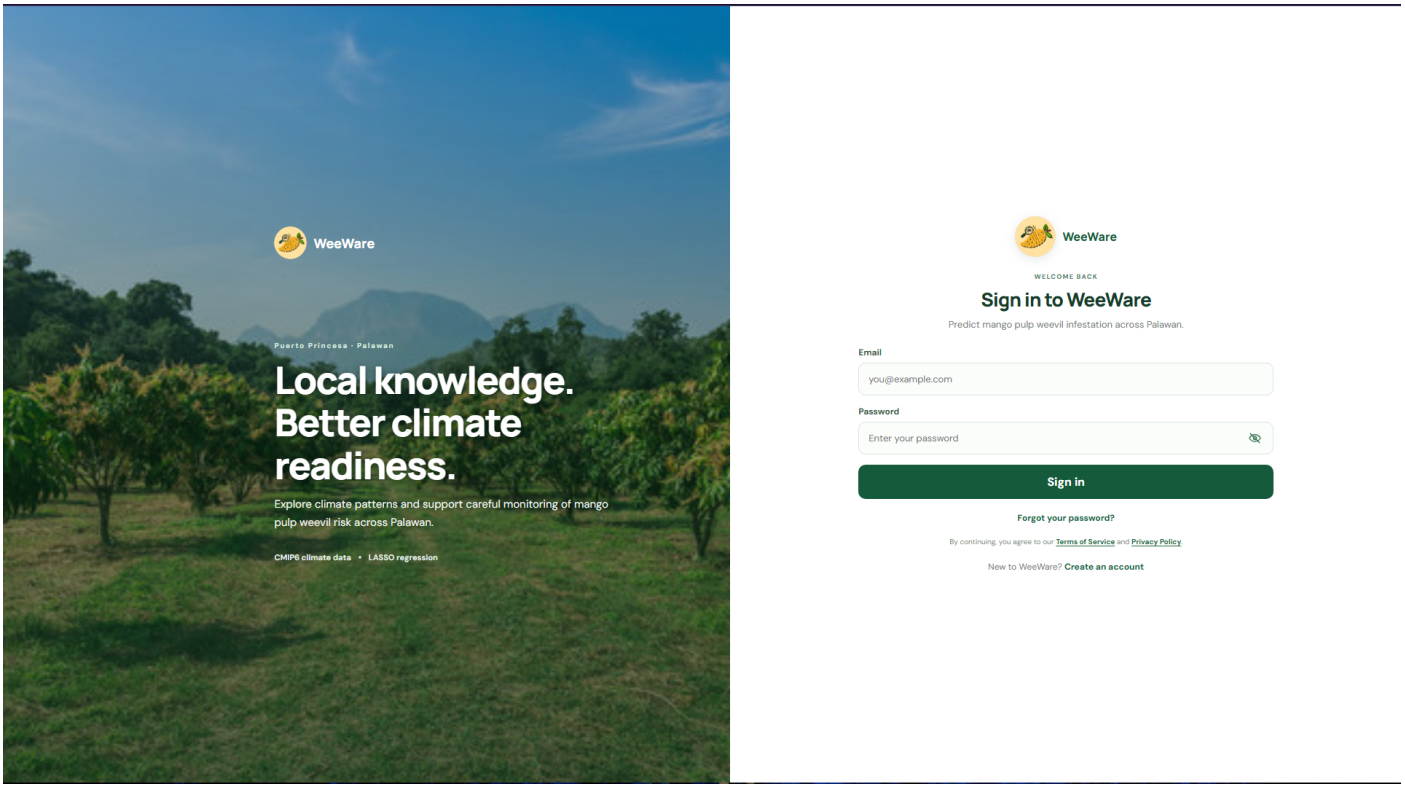
Client screenshot: public home page



Public landing page and application overview.
File: screenshots/client-supplied/03-public-home.png

CLIENT-PROVIDED SCREENSHOT

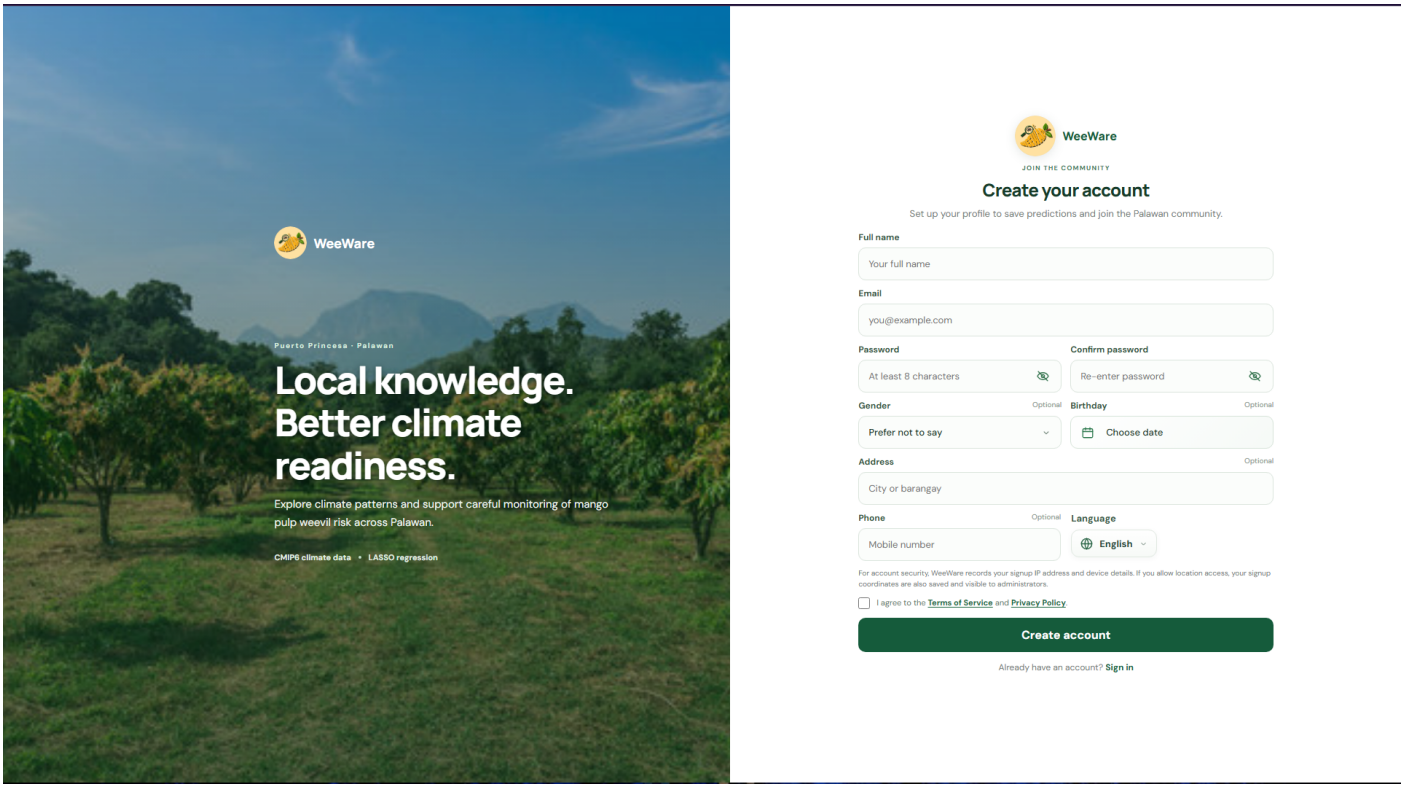
Client screenshot: sign in



Sign-in layout.
File: screenshots/client-supplied/04-sign-in.png

CLIENT-PROVIDED SCREENSHOT

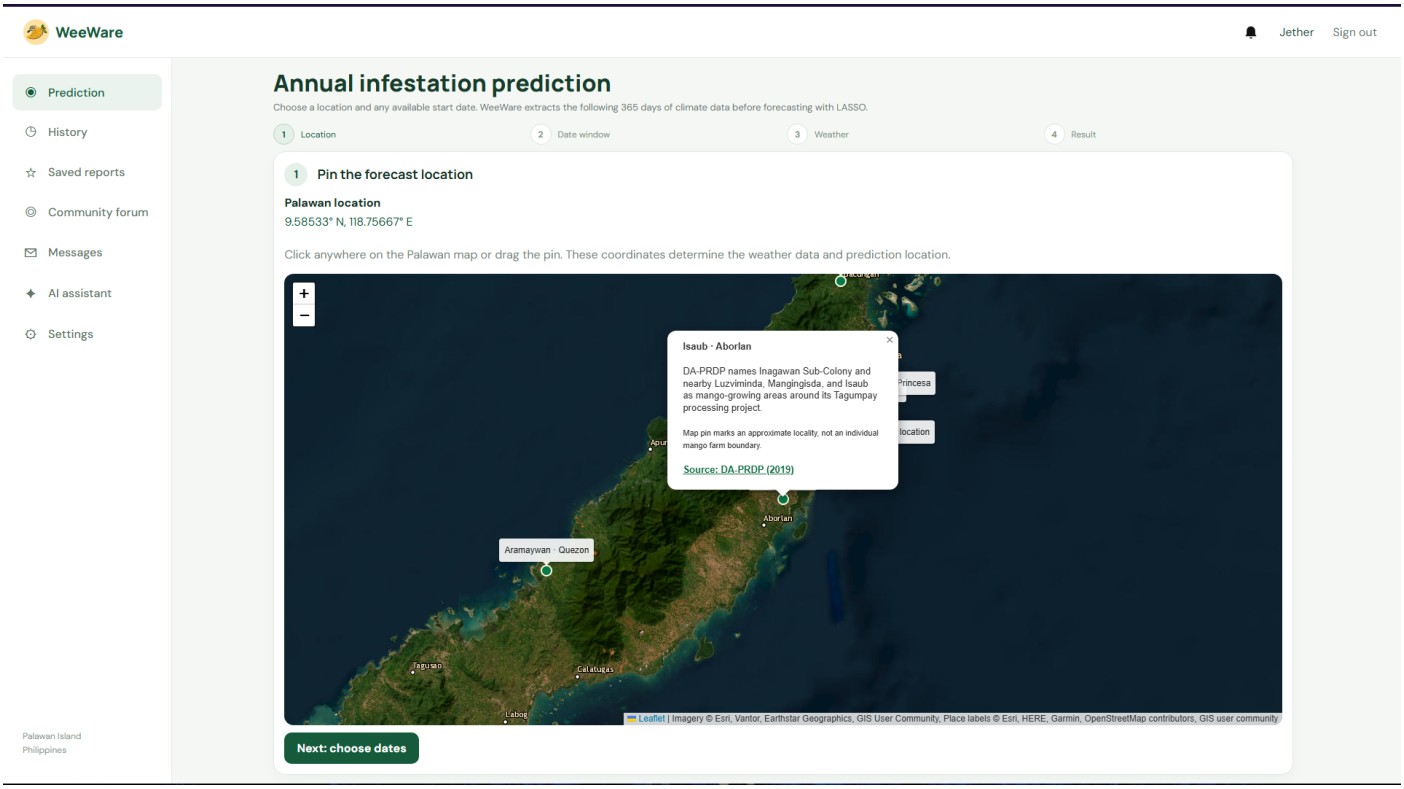
Client screenshot: create account



Signup form and account setup.
File: screenshots/client-supplied/05-create-account.png

CLIENT-PROVIDED SCREENSHOT

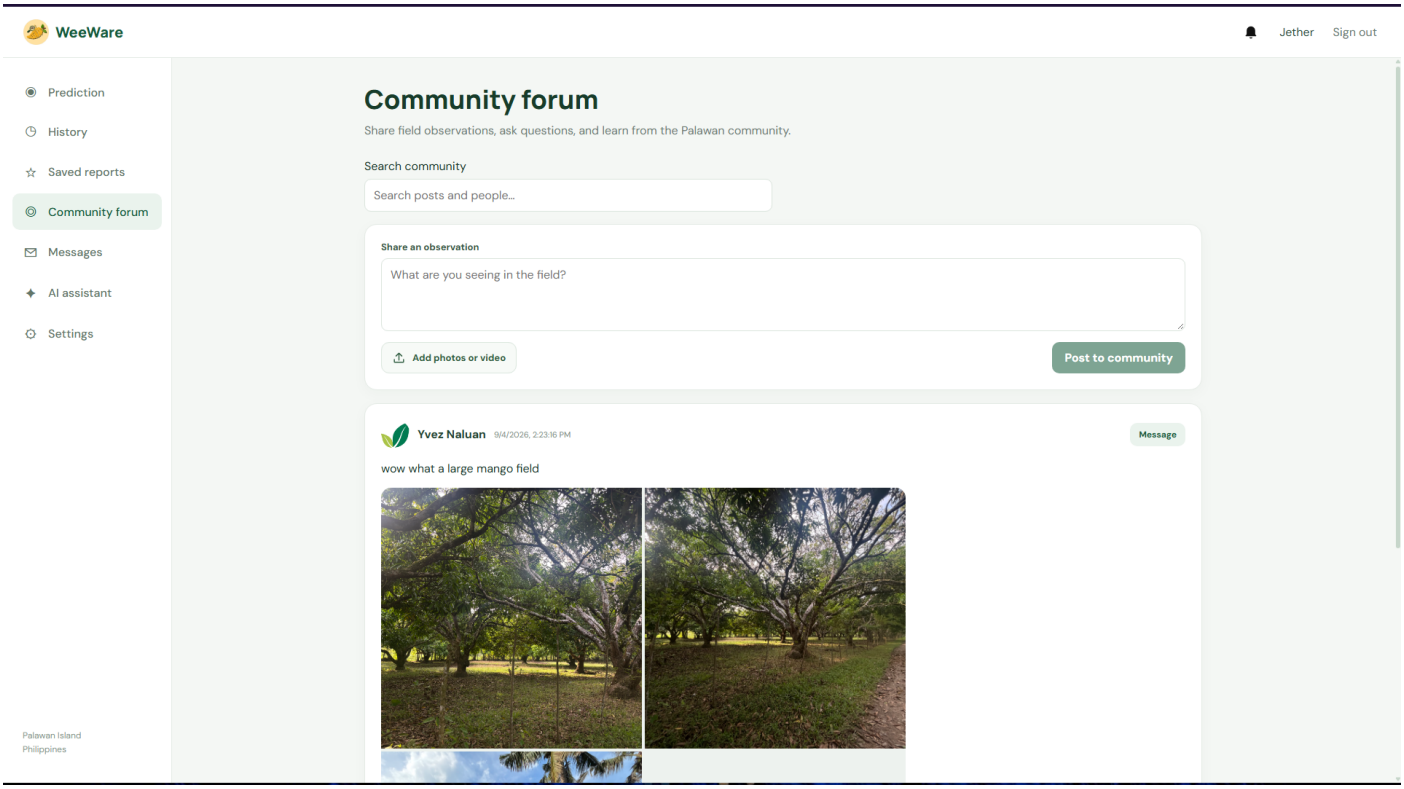
Client screenshot: location step



Palawan map, coordinates, and prediction wizard control.
File: screenshots/client-supplied/06-prediction-map-step.png

CLIENT-PROVIDED SCREENSHOT

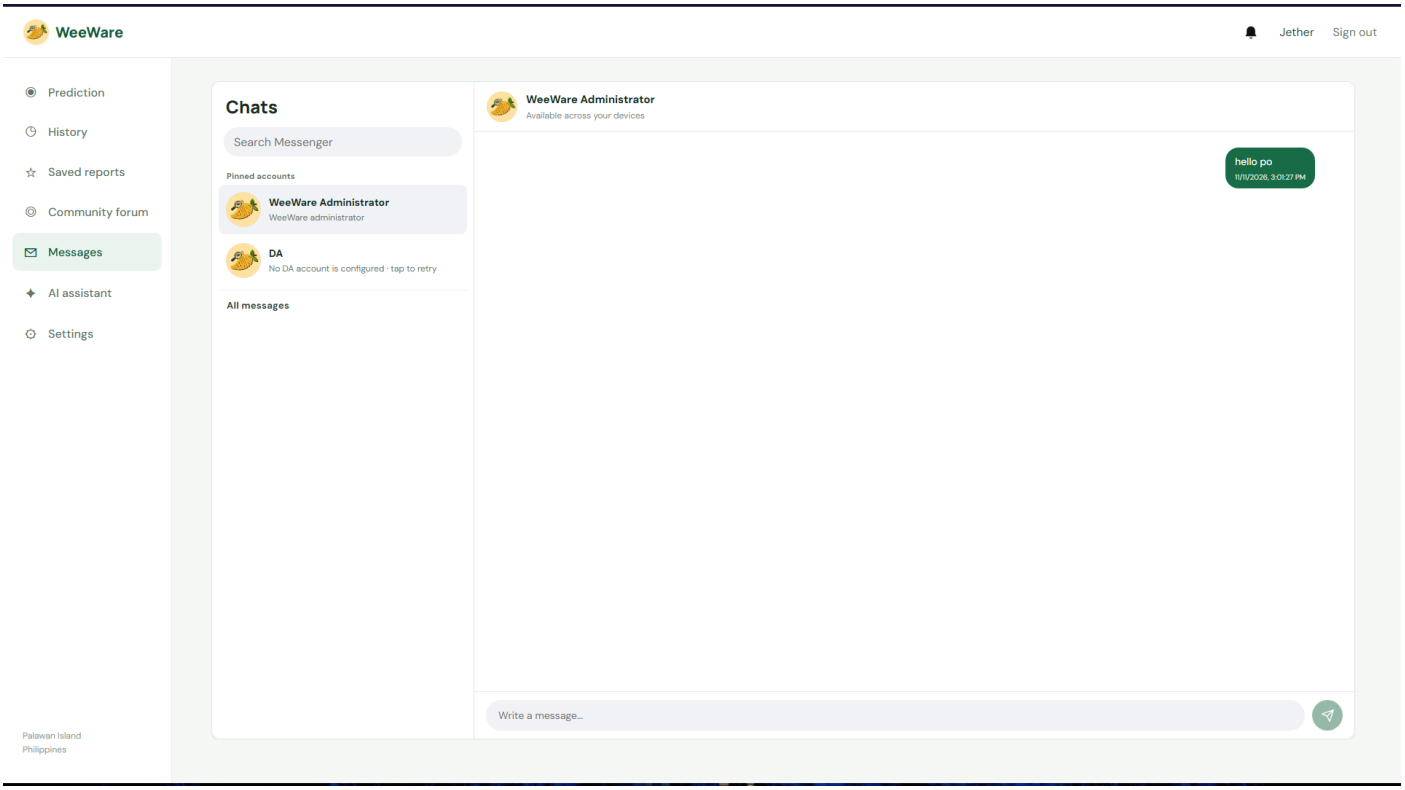
Client screenshot: community forum



Forum composition and post media.
File: screenshots/client-supplied/07-community-forum.png

CLIENT-PROVIDED SCREENSHOT

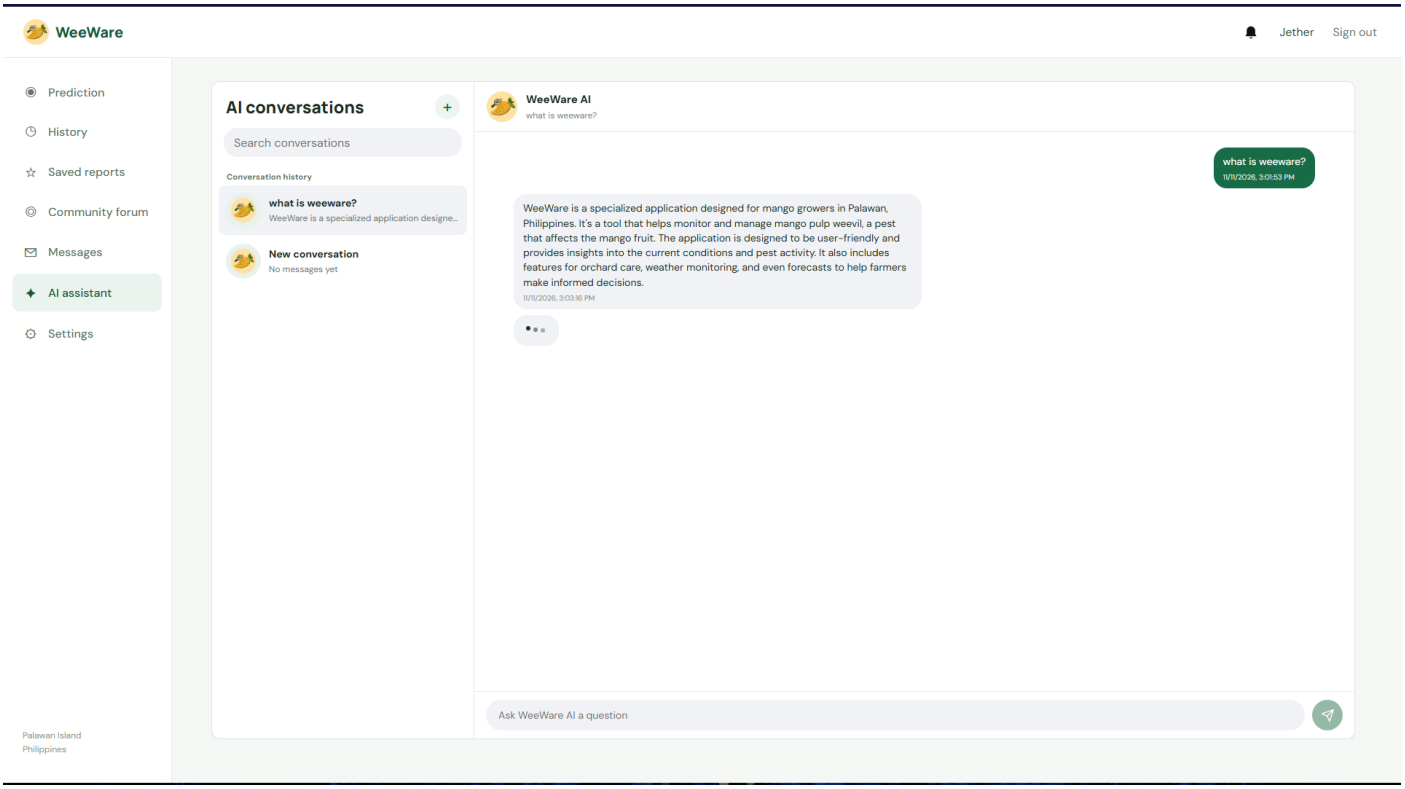
Client screenshot: messages



Group/direct message view.
File: screenshots/client-supplied/08-messages.png

CLIENT-PROVIDED SCREENSHOT

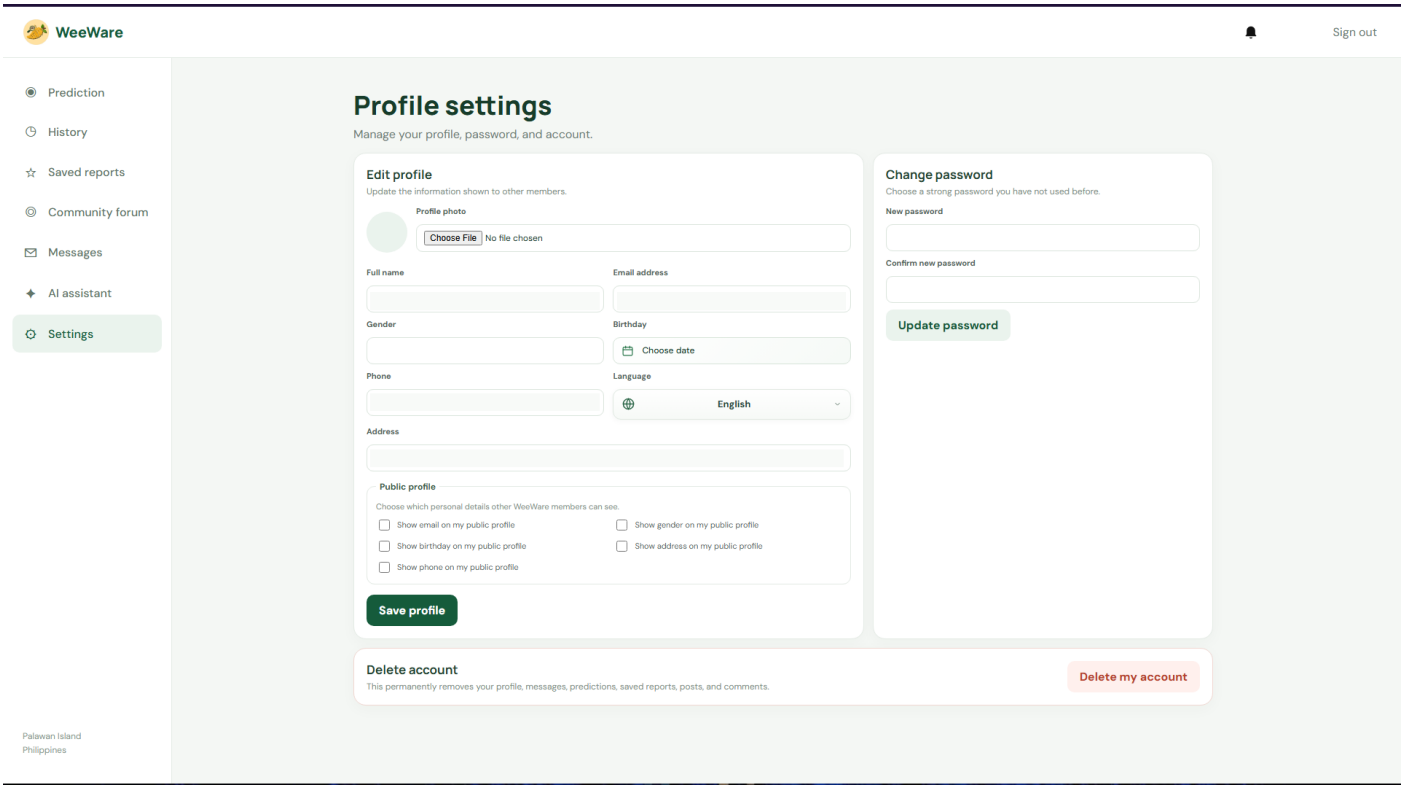
Client screenshot: AI assistant



Assistant conversation view.
File: screenshots/client-supplied/09-ai-assistant.png

CLIENT-PROVIDED SCREENSHOT

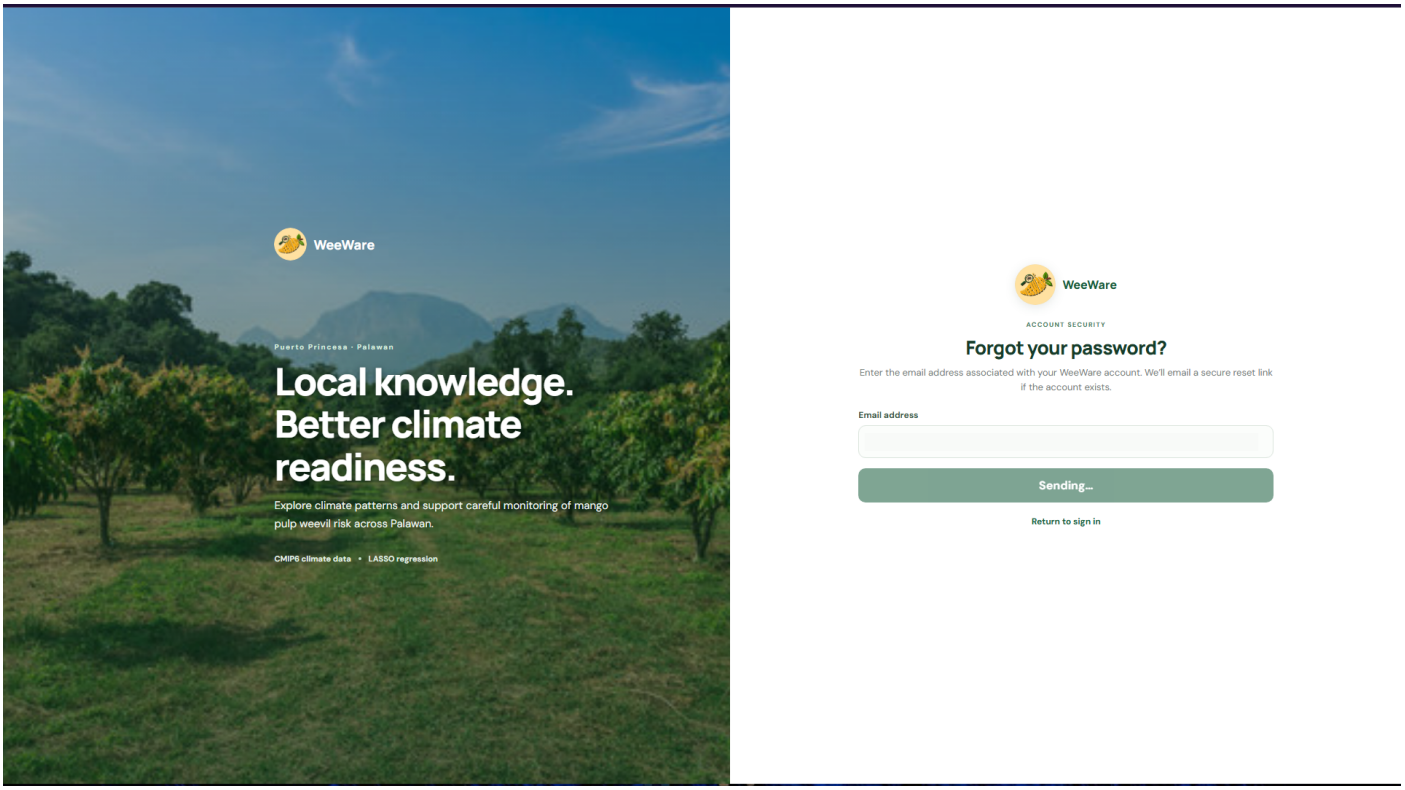
Client screenshot: settings (personal fields redacted)



Email, name, phone, and address values are masked in this report copy.
File: screenshots/client-supplied/10-settings-redacted.png

CLIENT-PROVIDED SCREENSHOT

Client screenshot: reset request (email redacted)



Shows the reset request in progress; it does not verify that email was delivered.
File: screenshots/client-supplied/11-forgot-password-redacted.png